Visual Justice

Racial Representation in Mugshot Coverage of U.S. Crime News

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Abstract

This paper studies the visual representation of crime in U.S. news and whether policies aimed at protecting the identity of arrested individuals affect crime reporting. We assemble 3.46 million crime articles published by 1,848 outlets between 2015 and 2024 and measure their images with a face-level pipeline that detects mugshots, infers race, and links the same individual across articles. Benchmarking against FBI arrest records, we show that Black arrestees are over-represented in mugshot coverage relative to their share of arrests, and the gap is concentrated in high-volume, low-severity offenses such as DUI and drug crimes, where publishing any given booking photograph is discretionary. We then exploit staggered state restrictions on mugshot disclosure in a difference-in-differences design. The bans reduce mugshot publication by about a third and compress the over-representation of Black arrestees where it was largest. Next, we investigate whether crime reporting also adjusts to the removal of mugshots. Using the same design, we show that outlets replace booking photographs with other images, cover fewer arrests and more investigations, and also write about crime differently. Finally, we provide preliminary evidence that reducing mugshot use affects crime perceptions by reducing the salience of crime.

1 Introduction

Over the past decade, news consumption has become increasingly visual. As social media became the primary information medium,[1] even outlets that traditionally relied on written reporting now

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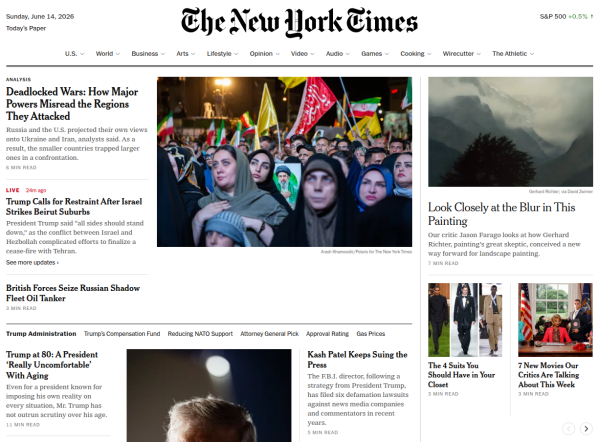
[1]See 2025 Digital News Report by the Reuters Institute.

Figure 1: New York Times Landing Page

(a) 2013



(b) 2026



rely extensively on images and videos, as illustrated in Figure 1. Work in psychology and, more recently, economics has shown that images are processed much faster than text (Thorpe, Fize and Marlot, 1996; Ciccarone, 2025) and carry information of their own, which can be slanted in ways the accompanying text is not (Caprini, 2024), raising questions about the consequences of this shift.

An area where the use of images could be particularly important is crime news. Crime stories are among the most widely consumed forms of news in the United States.[2] Existing research has shown that which crimes are covered and how they are reported affects beliefs about crime and criminal justice outcomes (Couttenier, Hatte, Thoenig and Vlachos, 2019; Philippe and Ouss, 2018; Ash and Poyker, 2024). However, this literature focuses almost exclusively on linguistic patterns, even when analyzing television coverage, and largely neglects the visual component of crime reporting (Moreno-Medina, Bayer, Ba and Ouss, 2025). This is despite recent evidence that visual features of suspects can be pivotal in shaping the judicial process (Ludwig and Mullainathan, 2024). Additionally, images convey information that text does not: an article can report an arrest without mentioning the race of the person arrested, while the photograph beside it shows it clearly. In light of the evidence on the persuasive impact of images (Peng, 2018; Caprini, 2024; Ciccarone, 2025) and on the effects of crime news (Lim, Snyder and Strömberg, 2015; Philippe and Ouss, 2018), who is depicted in crime coverage could be an important determinant of readers' perceptions of crime and their attitudes toward those accused of it.

In this paper, we fill this gap by studying the visual representation of arrestees in crime news

[2]Pew Research Center (2024).

and how it differs across racial groups. To do so, we focus on a distinctive visual element of U.S. crime reporting: the publication of mugshots. Mugshots are photographs taken shortly after an arrest as part of the booking procedure, capturing the suspect in a standardized format intended to facilitate identification. News organizations frequently obtain these booking photographs from law enforcement agencies and republish them alongside crime news, so police agencies supply the images while outlets decide whether to publish them. While police departments have been shown to post mugshots selectively along racial lines on social media (Grunwald, Nyarko and Rappaport, 2022), their circulation through news outlets has not been systematically analyzed.

We compile a dataset of 3.46 million U.S. crime articles published between 2015 and 2024 by 1,848 news outlets, together with the images published alongside them. We classify each image for the presence of a mugshot, infer the race of the person depicted, and cluster faces to link the same individual across articles. Mugshot publication is widespread but has declined over the sample period. White individuals account for the majority of those depicted, followed by Black individuals, with a long tail of other racial groups. Deduplicating to unique individuals narrows this gap, suggesting that Black individuals who appear in mugshot articles are reprinted less often than their White counterparts. We further show that they are covered by fewer outlets and for shorter periods. Mugshot coverage of Black arrestees is therefore broader because it spans more individuals, although each is covered less repeatedly.

Next, we investigate whether arrestees of different races are over- or under-represented in mugshot coverage relative to their share of arrests. Using arrest records from the FBI's National Incident-Based Reporting System (NIBRS), we compute for each racial group the ratio of published mugshots to arrests. We find that overall these ratios are small for both White and Black arrestees. However, comparing the two ratios, Black arrestees are over-represented relative to their White counterparts for both violent and non-violent crimes.

We then analyze the heterogeneity of this Black over-representation across states and crime types. First, while Black arrestees are over-represented in most states, the highest Black-to-White mugshot-to-arrest ratio is found among local outlets in western states. Second, White arrestees are over-represented in inherently newsworthy crimes, such as fraud and sexual offenses, while Black arrestees are over-represented in high-volume routine offenses such as DUI and drug crimes. Arrests in this second category are far more than what news outlets can cover, so publishing any given mugshot is likely to be a discretionary editorial choice rather than a response to the news value of the case, and are therefore the most likely to respond to any policy limiting the circulation of such

images.

After establishing these racial disparities in mugshot representation, we investigate whether they can be attenuated by reducing the use of mugshots. For this purpose, we study policies that restrict the disclosure of booking photographs by police agencies. In particular, we look at policies implemented by four states, New York, Oregon, Utah, and Louisiana, and leverage a staggered difference-in-differences design (Callaway and Sant'Anna, 2021) to identify the effects of these policies. We first show that these policies reduce mugshot publication in treated states by about a third, and by more in coverage of arrests and pre-trial proceedings, where booking photographs are most relevant.

Investigating the heterogeneity across states, we find that the effect is driven entirely by Utah and Louisiana and is statistically indistinguishable from zero in New York and Oregon. This may reflect several factors. First, states differ in how strictly agencies enforce the restriction. Second, states differ in the mix of crimes they report, and mugshot publication concentrates on the high-volume, low-severity offenses whose prevalence varies geographically. Third, if outlets have a strong preference for publishing mugshots, they may substitute local crime stories with stories from other states, for which they still have access to these images. Since we cannot rule out any of these channels, the results should be interpreted as intention-to-treat effects.

Having shown that the bans reduce mugshot publication, we then ask whether they also reshape its racial composition. For each racial group we construct a representativeness ratio, defined as the group's share of published mugshots divided by its share of arrests, so that a value above one means the group appears in mugshot coverage more often than its share of arrests would imply. We use this double normalization to account for the voluntary participation in NIBRS, which expanded sharply over our sample period. We estimate the effect of the bans on this ratio separately for Black and White arrestees, and take the difference between the two coefficients to capture changes in Black over-representation.

In Utah, the ban reduces the Black representativeness ratio by 0.282 while leaving the White ratio unchanged, narrowing the Black–White difference by three-quarters of the state's pre-ban gap. The effect is larger in coverage of arrests and pre-trial proceedings, where the Black ratio falls by 0.383, and the disparity narrows by slightly more than Utah's entire pre-ban difference. Louisiana moves in the same direction but with much smaller magnitudes. In the same sample, its Black ratio falls by 0.081, about a fifth of its pre-ban level, and the Black–White difference narrows by roughly half of the pre-ban level. Both states begin with Black over-representation well above the

never-treated average and converge toward it after the policy implementations, without reversing it.

Beyond racial composition, we ask whether the content of crime articles also changes when booking photographs are no longer available. Looking at the image composition, we find that outlets do not simply drop the image but replace mugshots with photos of other people and images that portray nobody. Additionally, they move coverage away from the arrest and pre-trial stages, where booking photographs originate, toward investigations and later stages of a case.

Next, we turn to analyze the narrative used by the articles with and without mugshots. Descriptively, we show that the text alone predicts the presence of a mugshot beyond the crime type and the procedural stage, with mentions of names of the people involved being the single most predictive feature. Additionally, we show that in articles with mugshots, the share of perpetrator-centered sentences is 12.56 percentage points higher than in articles without a mugshot. In the analysis, we use a summary measure of these reporting features by looking at the predicted probability that an article carries a mugshot, computed from its crime stage, its crime type, and its masked text as an outcome, and ask whether this changes with the bans. We show that, in Utah, this probability falls by 6.7 percentage points, about a third of its pre-ban level, and by around 2 percentage points within outlets. Decomposing the prediction into the contribution of the set of cases covered and the article's wording shows that these two components contribute about equally to the main effect, suggesting that Utah newsrooms covered different cases and wrote about them differently. In Louisiana, where the first stage is much weaker, we find more muted effects that go in the opposite direction: news outlets change their reporting, moving towards cases that typically carry a mugshot, while the language does not change.

Finally, we provide preliminary evidence on whether these influences readers. Exposure to crime news has been shown to raise concern about crime and support for punitive policy (Lim et al., 2015; Philippe and Ouss, 2018; Mastrorocco and Minale, 2018; Ash and Poyker, 2024), and mugshots make that coverage more vivid by attaching a face to it, so restricting them should reduce the salience of crime among readers. Using Google Trends, we investigate whether residents of treated states searched less intensively for crime-related queries after the bans. We find a decline of about 7 percent in crime-related search intensity, which does not appear for a set of placebo queries unrelated to crime. However, because search behavior is only a proxy for perceptions, we treat these results as preliminary.

This paper contributes to the growing literature in economics on slant and bias in *visual* media content. While a large literature documents partisan and ideological slant in news text (Gentzkow

and Shapiro, 2008, 2010; Durante and Knight, 2012; Djourelova, 2023), recent work has begun to measure bias in the images that accompany the news, showing that visual framing carries persuasive power of its own (Peng, 2018; Boxell, 2021; Caprini, 2024; Ciccarone, 2025). We contribute to these works in two ways. First, focusing on the supply-side choice of images we move beyond political slant and investigate the presence of racial disparities in images focusing on crime news, closer to Ash, Durante, Grebenshchikova and Schwarz (2021) who document over-representation of certain groups in the media. Second, we move beyond the experimental setting and provide novel evidence on the effects of limiting the use of these mugshots on such disparities in the context of crime news.

Additionally, we contribute to the long-standing literature on the racialized portrayal of crime. Content analyses of local and network television news consistently find that Black suspects are over-represented as perpetrators relative to arrest statistics and under-represented as victims, and are portrayed in more threatening and less individualized ways than White suspects (Entman, 1992; Gilliam, Iyengar, Simon and Wright, 1996; Dixon and Linz, 2000). A central theme in this work is that such disparities reflect editorial assessments of *newsworthiness* that are themselves grounded in racial and gender stereotypes, so that coverage decisions are most discretionary—and most racially skewed—precisely for routine, high-volume offenses rather than for inherently newsworthy crimes (Lundman, 2003). Our results showing that Black over-representation is higher for DUI and drug offenses while White arrestees are over-represented in more newsworthy crimes provide large-scale, image-based evidence consistent with this mechanism. We also go a step further, and investigate whether policy interventions are effective in reducing this racialized crime portrayal.

This paper also contributes to the literature on the downstream consequences of crime coverage for attitudes, beliefs, and institutional outcomes. Exposure to crime news raises concern about crime and demand for punitive policy (Lim et al., 2015; Philippe and Ouss, 2018; Mastrorocco and Minale, 2018; Ash and Poyker, 2024; Mastrorocco and Ornaghi, 2025), and racialized coverage in particular heightens stereotypical associations between race and crime that, in turn, increase support for punitive policies and shape behavior (Gilliam and Iyengar, 2000; Hetey and Eberhardt, 2014). Complementary work in psychology shows that the race–crime association operates through visual channels: Black faces facilitate the detection of crime-relevant stimuli, and more stereotypically Black features predict harsher sentencing in capital cases (Eberhardt, Goff, Purdie and Davies, 2004; Eberhardt, Davies, Purdie-Vaughns and Johnson, 2006). Because mugshots are often the only cue that conveys a suspect's race in coverage that is otherwise textually neutral, this strand motivates why the visual disparities we document may matter for real-world outcomes and further motivates

our analysis in the second part on the effects of mugshot bans. We also contribute to this literature directly, providing preliminary evidence that the salience of crime among readers decreases after mugshot publication was restricted.

Finally, we contribute methodologically and to the policy debate on mugshot disclosure. We develop a face-level pipeline that combines mugshot detection, image-based race inference, and face-embedding clustering, enabling individual-level analysis of news visuals at scale and extending the algorithmic-measurement tools recently introduced for studying media images and judicial decisions (Ash et al., 2021; Ludwig and Mullainathan, 2024). The closest prior evidence on selective racial diffusion of mugshots comes from Grunwald et al. (2022), who show that police-maintained Facebook pages over-report Black suspects relative to local arrest rates. We extend this analysis to the media market, and exploit state-level disclosure restrictions to provide causal evidence on how limiting mugshot availability affects racial representation. In doing so, we speak to an active policy and legal debate over the publication of booking photographs, which courts and newsrooms have increasingly restricted on grounds like presumption of innocence and reputational harm, but which has largely proceeded without systematic causal evidence.[3]

The remainder of the paper is organized as follows. Section 2 describes the data and methods, while Section 3 presents the results on the use of mugshots in news outlets and on racial disparities in such mugshots. Section 4 presents the analysis on the effects of mugshot bans, and Section 5 concludes.

2 Data and Methods

In this section, we introduce the data and how we extract variables from text (Section 2.1) and images (Section 2.2) for an analysis of mugshot prevalence in crime news and the representation of mugshots relative to arrests.

2.1 Data and sample constructions

We combine two datasets: a large corpus of online crime news and individual-level arrest records from the FBI's National Incident-Based Reporting System (NIBRS). The underlying news corpus is built from articles indexed by NewsAPI.ai. Our cleaned analysis window runs from January 2015 through September 2024. News articles in this dataset contain the article URL, publication date,

[3]See, e.g., *Houston v. Maricopa County*, 116 F.4th 935 (9th Cir. 2024).

outlet, byline, title, a text excerpt, and the URL of the main embedded image, which we use to scrape the image itself. From this raw collection we isolate articles that report on a specific crime incident.

To extract crime incidents, we train a binary classifier in three steps: First, a large language model (GPT-5-mini) labels a random sample of articles. Second, the sample is manually verified. Third, the labels are distilled into a lightweight model that is then applied to the full corpus. For these lightweight models, we select the performance-maximizing combination across three architectures (logistic regression, random forest, and XGBoost) and two input representations (word frequencies or term-frequency–inverse-document-frequency (TF–IDF)). The best performing model is an ensemble of XGBoost and logistic regression on TF-IDF features over the title and body, treating an article as a crime incident when either classifier fires. We repeat this pipeline to extract two additional variables on the filtered corpus: a multinomial classifier that initially distinguishes eight stages of the criminal procedure and a multi-label classifier that assigns each article to one or more of twenty crime types (homicide, assault, sexual offense, robbery, burglary, theft, drug offense, weapons offense, fraud, DUI, kidnapping or trafficking, public order, and others). For validation and broad descriptive use, we aggregate the eight stage predictions into four categories: investigation; arrest or charging; court proceedings (pretrial, trial or verdict, and sentencing); and post-case proceedings (corrections or release, appeal, and aftermath). The original labels remain available when an analysis specifically isolates arrest and pretrial coverage. Appendix Tables A.1 report validation against the human annotations, while the held-out teacher-label metrics measure the separate task of reproducing the LLM labels.

We further restrict the sample to outlets with at least twenty-five unique-image crime articles in the corpus, in order to remove very thinly covered outlets. The qualifying export contains 6,060,469 crime-incident records. After applying the language, news-site, date, and duplicate filters described in Table 1, the main analysis dataset contains 3,463,137 articles from 1,848 distinct outlets, of which 2,013,928 have an image processed by the face pipeline.

Outlet attributes are obtained from a separate GPT pass that classifies each outlet along three dimensions: medium (newspaper, TV, radio, website, magazine, wire service), editorial scope (news vs. non-news), and geographic scope (local vs. national). We identify each outlet's headquarters city, county, and state by a GPT-conducted web search. Of the 1,848 outlets, 1,103 are classified as local, 728 as national, and 17 have unresolved geographic scope. The sample contains 658 online-only websites, 574 newspapers, 304 TV stations, 182 radio outlets, 119 magazines, and 11 wire services.

Table 1: Descriptive overview of the analysis dataset.

Articles, images, and mugshots	**Count**
Qualifying crime-incident export	6,060,469
Main descriptive analysis sample	3,463,137
with an image processed by the face pipeline	2,013,928
with at least one mugshot face	261,153
Unique mugshot individuals (face clusters)	185,399
classified as Black	63,830
classified as White	78,257
Outlets in the main sample	
Total outlets	1,848
local	1,103
national (non-local)	728
other or unresolved scope	17
Outlets with at least one mugshot article	1,526
Articles per outlet	
Mean	1,874
Median	470
25th–75th percentile	132–1,680

Notes. The qualifying export contains crime incidents from outlets with at least 25 unique-image crime articles. The main descriptive sample additionally restricts to English, nonduplicate news-site articles published from 2015-01-01 through 2024-09-30. An image is counted when its mapped image was processed by the face pipeline. Mugshots use the adopted face-level probability threshold of 0.9. Individual counts use the adopted Chinese-Whispers identity clusters and include only clusters observed in the main descriptive sample.

The arrest data comes from the FBI's National Incident-Based Reporting System (NIBRS). While participation and data disclosure are voluntary at the agency level, a federal transition deadline at the start of 2021 has accelerated adoption. The NIBRS covered population increases by 90 percent between 2019 and 2024, resulting in an 84 percent coverage of the total US population in 2024. NIBRS records are reported at the individual arrest level, with offense, race, sex, age, agency identifier, and date. We map agencies to states and counties using the FBI's agency reference file.

2.2 Image variable extraction

We analyze the downloaded main images of crime news articles to identify whether the image contains a mugshot. Because an image can contain multiple faces, we first extract every face crop from every crime-article image using a face-detection model (MTCNN) (Zhang et al., 2019). We

identify mugshots by having a vision-language model label a sample of face crops as mugshot vs. not, manually validate the sample, and then fine-tune an image classification model (EfficientNet-B0) on those labels (Tan & Le, 2019). We adopt a conservative face-level probability threshold of 0.90 to prioritize precision. An image is recorded as a mugshot if any of its detected faces exceeds this threshold.[4]

For each detected face we infer race and gender using a per-face classifier. The classifier leverages embeddings from the same face-recognition model and feeds them into a small, fully-connected neural network trained on the FairFace dataset of race- and gender-annotated face crops. The model returns one of six race categories (White, Black, Asian, Indian, Latino-Hispanic, Middle Eastern) and a binary gender label. In the representation-by-crime analysis, each mugshot face in each published article is one impression: repeated publication of the same person is retained, and an article containing multiple mugshot faces contributes one observation per face.

Finally, since mugshots of the same individual can appear in many articles, either as the same image republished across outlets or as different mugshot variants of the same person, we identify duplicate faces on the basis of facial characteristics. To recover unique individuals, we represent each mugshot face by its FaceNet embedding, build a k-nearest-neighbor graph over the full set of detected faces using cosine similarity, and apply the Chinese-Whispers community-detection algorithm at a high similarity threshold of 0.90. We manually verify that clusters represent the same individual on a random sample and calibrate the similarity threshold by inspecting near-threshold image pairs. We attribute a cluster to the year-month of its first mugshot appearance and use it as the unit of observation whenever the question concerns people rather than impressions.

2.3 Language variable extraction

In addition to the image analysis, we use article text to examine how strongly the written presentation of a crime predicts the accompanying visual presentation. First, we test whether the article's main image can be predicted from the text. To limit direct label leakage, we preprocess the headline and the first 2,000 characters of the article. Named-entity recognition replaces names of people, organizations, locations, and remaining proper nouns with placeholders. Similarly, dictionary rules replace explicit race and gender terms. We also remove direct visual references—such as *mugshot* or *booking photograph*— and replace years, month names, numerical dates, and times with a common

[4]The face-level mugshot classifier is fine-tuned on 1,500 mugshot and 1,500 non-mugshot face crops. At the production threshold of 0.90, held-out test performance ($n_{\text{test}} = 300$) is: accuracy 0.920, precision 0.984, recall 0.848, and F1 0.911.

calendar tag. To illustrate, the stylized sentence "The Black male suspect John Smith was arrested in New York in May 2021" becomes "The *RACE GENDER* suspect *PERSON* was arrested in *LOCATION* in *CALENDAR CALENDAR*." Generic tags are retained as the presence of references to people and institutions may themselves be features of reporting style, while their concrete identities are discarded.

For the mugshot prediction, we estimate three regularized logistic classification models. The *incident model* uses only extracted text features, in particular the crime stage, crime type, and their interaction. The *Title and excerpt model* uses the masked text represented by TF-IDF-weighted bigrams. The *combined model* adds the masked-text contribution to the incident model's predicted log-odds. We map each model's scores into calibrated probabilities using a separate calibration sample. The eligible model-development pool contains approximately 1.13 million articles, 16.5% of which contain a detected mugshot. The sampled development data are divided into approximately 70% training, 15% calibration, and 15% calibration-evaluation observations. To avoid learning changes induced by the mugshot-restricting policies introduced in Section 4, this pool includes articles from never-treated states throughout the sample period but only pre-policy articles from New York, Utah, Oregon, and Louisiana. We additionally evaluate transport performance on 20,328 articles published in 2024 by 77 outlets that do not appear during model training.

We separately measure how much relative weight an article puts on the core actors involved in a crime: perpetrators and victims. We define a sentence as perpetrator- or victim-centered when it primarily develops a perpetrator's or victim's identity, biography, perspective, experience, behavior, legal status, or consequences of the crime [5]. A sentence is coded as mixed when it gives substantial attention to both sides and as other when it centers neither side, for example by discussing police activity, court procedure, or journalistic context. We again construct the training labels through a similar model distillation pipeline. An LLM assigns the narrative-focus states to a stratified sample of 6,473 crime-news sentences. We then fine-tune a contextual transformer, DeBERTa-v3-base, to reproduce these labels. For each target sentence, the deployed model receives the headline and up to three preceding and three following sentences as inputs and outputs two binary classifications: one for being victim- and one for being perpetrator-centered. Unlike a bag-of-words model, the transformer can use the semantic context to resolve whom a description concerns. We average the predicted probabilities from two trained models with different seeds and hyperparameters and

[5]The term *Perpetrator* here denotes a role in the discourse and should not imply anything about the individual's actual guilt.

select the decision thresholds using validation data. We measure performance on held-out GPT labels and evaluate the final classifier against a blinded human annotation of randomly sampled corpus predictions. We report both forms of validation, including class-specific precision, recall, and F1, in Appendix Table A.2. Because the mixed class is rare in the human-validation sample, its performance is reported separately and interpreted cautiously.

3 Mugshot Use in News Outlets

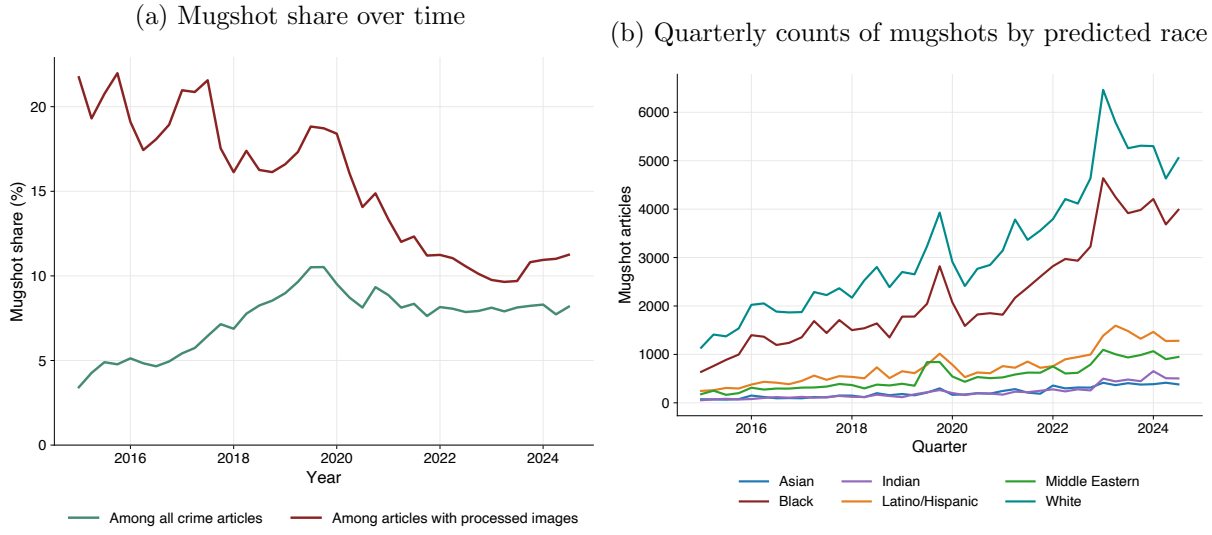
In this section, we present descriptive results on the use and racial composition of mugshots in news articles at the image and individual levels (Section 3.1) and on the racial representation of these images relative to NIBRS arrests (Section 3.2).

3.1 Mugshot prevalence and racial composition

We start by investigating the prevalence of mugshots in our corpus. Figure 2a shows the share of mugshots among crime articles using two different samples. On the one hand, it shows that the number of mugshots divided by the total number of crime articles has been increasing slightly over time, while on the other hand, the share of mugshots among articles with images went from roughly 19–22 percent in 2015 to roughly 11 percent in 2024 (Figure 2). This difference likely reflects the availability of articles with images in our sample. Appendix Figure A.3 shows that the number of crime articles has grown significantly over the 2015–2024 window, and the number of articles with at least one image follows a similar trend but increases more strongly, resulting in a rising share of image coverage over time. This is partly because more articles publish associated images, but mainly because older images have been taken offline or can no longer be collected. This does not mechanically bias the reported trend or subsequent policy effect estimation: attrition affects the composition of the observed images only if mugshots are systematically more or less likely to remain accessible than other image types. Most image loss results from broken links, expired hosting, and website migrations that operate at the page or outlet level rather than according to image content.

Decomposing these mugshots by the predicted race of the depicted individual shows that White individuals are the majority throughout the full period, followed by Black individuals. Latino-Hispanic, Middle Eastern, Asian, and Indian arrestees each account for substantially smaller shares, with no clear trend over the period (Figure 2b). In what follows, we therefore focus on the differences between the depiction of Black and White individuals.

Figure 2: Mugshot prevalence and race composition



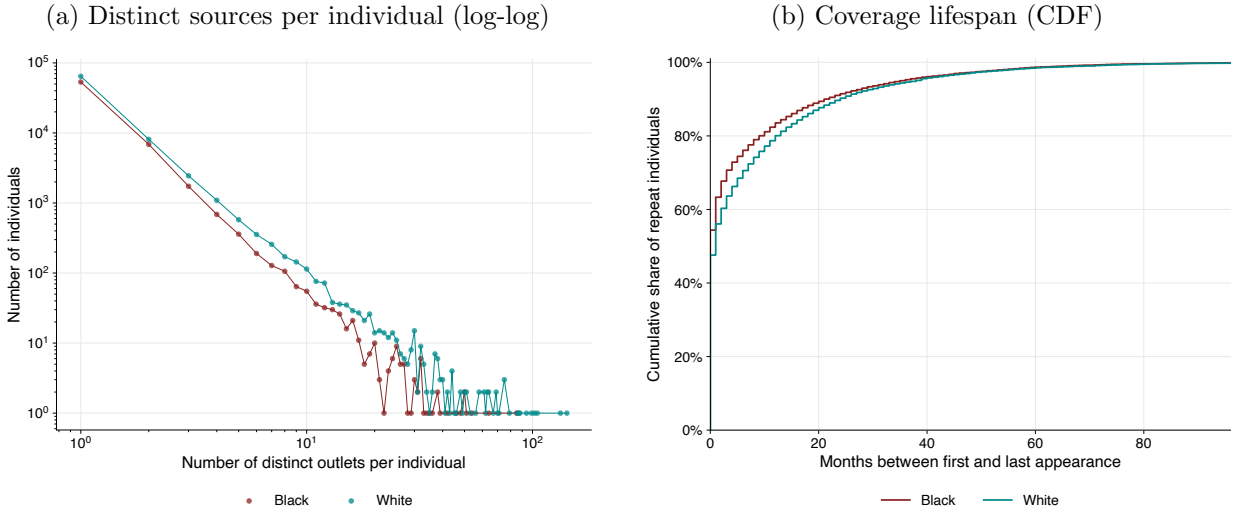
Notes. Trends in mugshot publications over time. Panel A plots the share of images that were classified as containing at least one mugshot over all crime articles, and all articles with images. Panel B plots the total number of articles with at least one mugshot separately by predicted race. Mugshots and races were obtained via face-level, custom-trained classifiers (training procedure detailed in Section 2.2). The race/ethnicity classes result from the annotations of the FairFace training dataset.

Next, we move from all mugshot images to unique individuals, obtained via the face-clustering procedure. Comparing the total impression with individuals reveals that each Black arrestee is reported less often than White ones (Appendix Figure A.4). This is explained by two complementary patterns. First, Black individuals are reprinted across fewer distinct outlets, as suggested by the heavier right tail of the distribution of distinct sources per White individual relative to Black (Figure 3a). Second, the time elapsed between an individual's first and last mugshot appearance—the coverage lifespan—is systematically shorter for Black individuals (Figure 3b). White individuals in mugshots, in other words, are depicted more frequently across both outlets and time, which mechanically inflates the White share among all mugshot images.

3.2 Representation relative to arrests

Next, we investigate whether the representation of Black and White arrestees in mugshots published in news articles matches their actual arrest rate using NIBRS data. We focus on 2022–2024, where NIBRS coverage is broadest, and compare the race composition of mugshot face impressions with

Figure 3: Individual-Level Mugshot Coverage Patterns by Race



Notes. The plot shows two aspects of the repeated re-publication of mugshots. Panel 3a depicts the distribution of the number of distinct outlets an individual mugshot arrestee occurs in by race (log-log transformed). Panel 3b shows the cumulative distribution function of the time difference between first and last occurrence by race.

that of NIBRS arrests.[6] [7] Appendix Figure A.8a shows that the race composition of arrestees is broadly similar to that of mugshot subjects, with White and Black individuals jointly accounting for the vast majority of arrests.

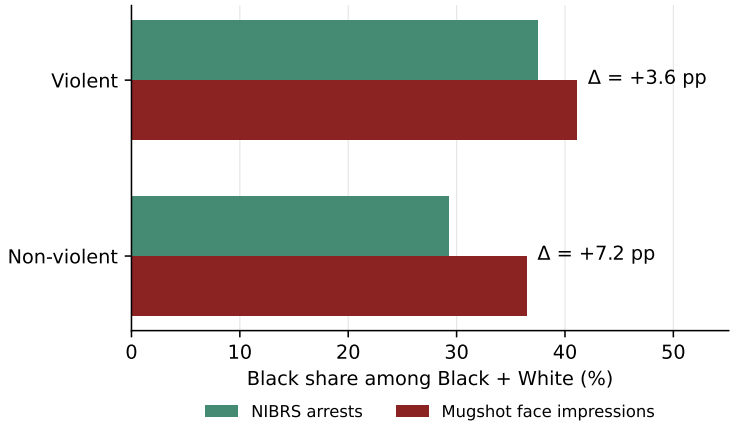
When comparing the mugshot-to-arrest ratio between Black and White arrestees (Appendix Figure A.8b), we see that Black individuals are over-represented in mugshot coverage relative to their share of arrests. The Black mugshot-to-arrest ratio substantially exceeds the White ratio in every year of the window, and the gap increases from 3.08 in 2022 to 3.97 in 2024.

Next, the over-representation is geographically heterogeneous. Dividing the Black mugshot-to-arrest ratio by the White ratio yields a representation index that is larger than one if Black arrestees are over-represented in mugshots relative to Whites. Appendix Figure A.5 reveals that this index is largest in the western states and smaller in the south and parts of the Midwest. Finally, the over-representation is present for both violent and non-violent crimes but is substantially larger for the latter, as shown in Figure 4. Decomposing the gap by individual crime type reveals that the over-representation is especially pronounced for low-severity offenses such as driving-under-the-influence

[6] We keep the repeated publication of the same person in this comparison and assign each impression to the state of the local outlet publishing the article.

[7] NIRBS reports two broad categories of arrests (group A and group B). Group A are more severe crimes that agency must report even if no arrest has been made while group B only includes crimes that need to be reported when an arrest was made. For this analysis we restrict to actual arrests in group A and combine data for group A an B.

Figure 4: Black vs White representation in mugshots by crime type vs arrests.



Notes. Share of Black faces in mugshots reporting on violent vs non-violent crimes compared to Black share in arrest data. The sample is restricted to 2022–2024 articles and arrests. An article is violent if any of the classified crime label is violent and nonviolent only if all qualifying labels are nonviolent.

Table 2: Predicting Mugshot Publication from Incident Characteristics and Story Text

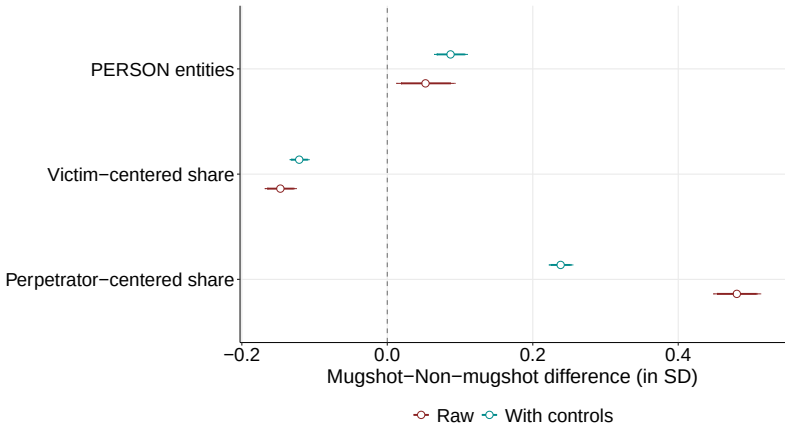
Model	Threshold	Avg. precision	Bal. accuracy	Precision	Recall	F1	Brier
Incident only	0.208	0.228	0.697	0.215	0.761	0.335	0.097
Title and excerpt only	0.250	0.325	0.716	0.309	0.614	0.411	0.096
Combined	0.264	0.326	0.719	0.310	0.621	0.413	0.094

Notes. Performance of calibrated logistic-regression models predicting whether the article's featured image is a mugshot. "Incident only" uses crime stage, crime type, and their interaction. "Title and excerpt only" uses TF–IDF features from the masked title and article excerpt. "Combined" uses both sets of predictors. Classification thresholds maximize F1.

and drug offenses (Appendix Figure A.9).

This pattern suggests that the visual over-representation is not mechanically driven by the racial composition of violent-crime arrests. Instead, it points to a substantive difference in the publication practices of mugshots that can be interpreted as a disparate judgment in what is considered newsworthy. In this interpretation, the same crime type is considered more newsworthy if an arrestee is Black, and this interacts with where the crime occurs. However, where this racial skew is rooted is not ex-ante clear. Previous work on crime reporting has shown that news outlets often copy or only mildly adapt information provided by law enforcement (Moreno-Medina et al., 2025). Whether changes in local information supply by law enforcement affect mugshot publication is therefore the focus of the following section.

Figure 5: Differences in Textual Characteristics Between Mugshot and Non-Mugshot Articles



Notes. Descriptive difference between articles with and without a mugshot as the main image for each outcome in the cleaned main analysis sample, divided by that outcome's pooled article-level standard deviation. Adjusted estimates include outlet, calendar-month, crime-stage, and crime-type fixed effects. Whiskers show 95% confidence intervals, with standard errors clustered by outlet.

3.3 Mugshot use and article text

The next step in understanding the reporting dynamics of mugshot publication is to assess whether using these images affects other article characteristics. To investigate this, we look at whether the written article predicts the presence of a mugshot. Table 2 reports the performance of the three models described in Section 2.3 on a source-held-out test sample of 20,328 articles from 77 outlets that do not appear in the training data. Adding the masked article text to the incident characteristics increases average precision from 0.228 to 0.326 and F1 from 0.335 to 0.413, while reducing the Brier score from 0.097 to 0.094. Because mugshots appear in only 11.6% of the test articles, a random ranking would have an expected average precision of 0.116. The improvement therefore indicates that the written presentation contains meaningful predictive information beyond crime type and procedural stage.

Among the masked textual features, the placeholder that replaced names of persons (`NER_PERSON`) has the largest positive coefficient in both the text-only model and the textual component of the combined model, as Figure A.1 shows. References to named persons are therefore particularly predictive of mugshot publication. This result does not, however, reveal who is mentioned or what role they occupy in the story. Therefore, we analyze the relative weight of the two major actor types involved in a crime: the person(s) committing it and the person(s) suffering its consequences.

Figure 5 presents the article-weighted comparisons of perpetrator- and victim-centered sentences (Appendix Figure A.2 shows the comparison in non-normalized values). While both mugshot and

non-mugshot articles relatively emphasize the perpetrator compared to the victim, this difference is significantly more pronounced in mugshot articles, where the share of perpetrator-focused sentences is, on average, 0.48 standard deviations higher. After controlling for outlet, calendar-month, crime-stage, and crime-type fixed effects, the difference remains 0.24 standard deviations. The corresponding adjusted difference in the share of victim-centered sentences is negative but less stark in absolute terms (-0.12 standard deviations).

Taken together, the predictive and descriptive results suggest that mugshot publication is associated with a broader form of perpetrator-centered reporting: the image foregrounds the depicted person, while the accompanying text devotes relatively more attention to the perpetrator and less to the victim.

4 Effect of Banning Mugshots

In this section, we investigate the effect of restricting mugshot publication on racial disparities in mugshot coverage. In particular, Section 4.1 describes the empirical strategy and Section 4.2 reports the results.

4.1 Empirical Strategy

Our identification strategy leverages the effects of policies restricting the publication of mugshots adopted at different times in the states of New York, Oregon, Utah, and Louisiana (see Appendix Table A.3). These policies require law enforcement agencies to stop releasing booking photographs unless the release serves a "specific law enforcement purpose", such as recognizing a fugitive. Appendix Figure A.7 shows that the mugshot share in treated states falls compared to control states around the adoption date. In most cases, however, treated and control states already diverge before the policy was implemented, consistent with outlets and agencies anticipating a policy that has already been approved. Therefore, in our analysis, we use the month of approval as the treatment date to account for this anticipatory response.

To isolate the causal effect of these policies, we leverage a difference-in-differences design (DID) with staggered treatment. Namely, we compare the share of mugshots ($MugshotShare_{st}$, in each month t and state s) across states affected by these policies and those that are not, before and after implementation. We run the following regression

$$MugshotShare_{st} = \beta Ban_s \times Post_t + \alpha_s + \alpha_t + \varepsilon_{st} \tag{1}$$

Where Ban_s equals one if state s adopted a mugshot policy and $Post_t$ equals one in and after the month of the policy approval, α_t and α_s are month and state fixed effects, and standard errors are clustered at the state level. Because adoption is staggered, we also report the estimator of Callaway and Sant'Anna (2021) to account for negative weights and contamination from already treated units that arise in a two-way fixed effects design with heterogeneous timing (Goodman-Bacon, 2021). We estimate the same specification at the article level, include outlet fixed effects, and therefore identify the effect within outlets rather than across them.

We then ask whether the policies changed the racial composition of the mugshots that continue to be published. For race $G \in B, W$, we define the Representation Ratio (RR)

$$R^G_{st} = \frac{\text{Mugshots}^G_{st}/\text{Mugshots}_{st}}{\text{Arrests}^G_{st}/\text{Arrests}_{st}},$$

as the share of crime images in state s and month t that are mugshots of individuals of race G, divided by race G's share of arrests in that state and month. The measure compares how heavily a group is represented among published mugshots with how heavily it is represented among arrests, so a larger R^G means more mugshot coverage per arrest. We use this double-normalized ratio rather than the simple mugshots-to-arrest ratio, $\frac{\text{Mugshots}^G_{st}}{\text{Arrests}^G_{st}}$, because compliance in sharing data to NIBRS is voluntary and displays a sharp increase during our sample period. As described in Section 2.1, the covered population grows by 90 percent between 2019 and 2024, with reported arrests doubling between 2021 and 2022 in more than half of all states. A measure with the simple arrest count in the denominator would therefore fall as a state's reporting coverage grew, and any change detected after the policy would be driven by the state's NIBRS onboarding rather than the mugshot restriction. Using the representativeness ratio allows us to account for this state-specific program expansion across years, assuming that agencies entering NIBRS do not differ systematically from those already reporting in the racial composition of their arrests.

To estimate the effect of mugshot-banning policies on such representation ratios, we estimate the same difference-in-differences regression we used in the first stage separately by race

$$R^G_{st} = \beta^G Ban_s \times Post_t + \mathbf{X'}_{st}\delta + \alpha_s + \alpha_t + \varepsilon_{st} \tag{2}$$

Table 3: Effect of Mugshot Bans on Mugshot Publication

	Mugshot share					
	Aggregate panel			Article level		
	Stacked	CSA	CSA Dyn.	Stacked	Stacked	CSA
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. All crime stages						
Mugshot ban × Post	−0.065***	−0.051	−0.057	−0.041***	−0.037**	−0.033
	(0.018)	(0.044)	(0.038)	(0.011)	(0.018)	(0.023)
Observations	6,822	3,120	3,120	2,235,976	2,235,976	4,047
Mean dep. var.	0.160	0.167	0.167	0.137	0.137	0.127
Panel B. Arrest and Pretrial						
Mugshot ban × Post	−0.104***	−0.140***	−0.132***	−0.065***	−0.061***	−0.055**
	(0.027)	(0.047)	(0.044)	(0.018)	(0.023)	(0.027)
Observations	6,734	2,738	2,738	833,113	833,113	2,831
Mean dep. var.	0.248	0.249	0.249	0.226	0.226	0.212
State FE	X	X	X	X	X	
Month FE	X	X	X	X	X	X
Outlet FE					X	X

Notes. Effect of mugshot bans on mugshot publication, measured as the share of images that are mugshots. Columns 1 to 3 report the results using an aggregated panel at the state level, while Columns 4 to 6 report the results using the article-level data to allow for outlet fixed effects. Columns 1, 4, and 5 report the results using a two-way fixed effects regression, while columns 2, 3, and 6 use the estimation by Callaway and Sant'Anna (2021). All columns include month fixed effects, columns 1 to 5 also include state fixed effects, and columns 5 and 6 include outlet fixed effects. Panel A reports the results for all articles, while Panel B only for articles about crimes at the arrest or pre-trial stage. Standard errors are clustered at the state level (in parentheses): * p < 0.1, ** p < 0.05, *** p < 0.01.

The regression includes state and month fixed effects and standard errors clustered at the state level. We also control for the share of each crime type reported in that state and month, $\mathbf{X'}_{st}$.[8]

To summarize the effect on racial disparity, we look at the difference $\beta^B - \beta^W$, which we estimate directly on $R^B_{st} - R^W_{st}$ to construct accurate standard errors. A negative β^B larger in magnitude than β^W means the ban reduced Black representation more than White representation, narrowing the gap between them. Conversely, an opposite pattern would suggest that such disparities in racial representation widened following the policies.

4.2 Results

4.2.1 First Stage: Mugshot Bans and Mugshot Publication

We start by estimating the effect of mugshot bans on the share of mugshots published in news outlets. Table 3 reports the first stage results pooling all treated states. Because adoption is staggered, a

[8]We report reduced-form estimates only. A two-stage least squares specification would violate the exclusion restriction since these policies plausibly affect television and other media alongside newspapers.

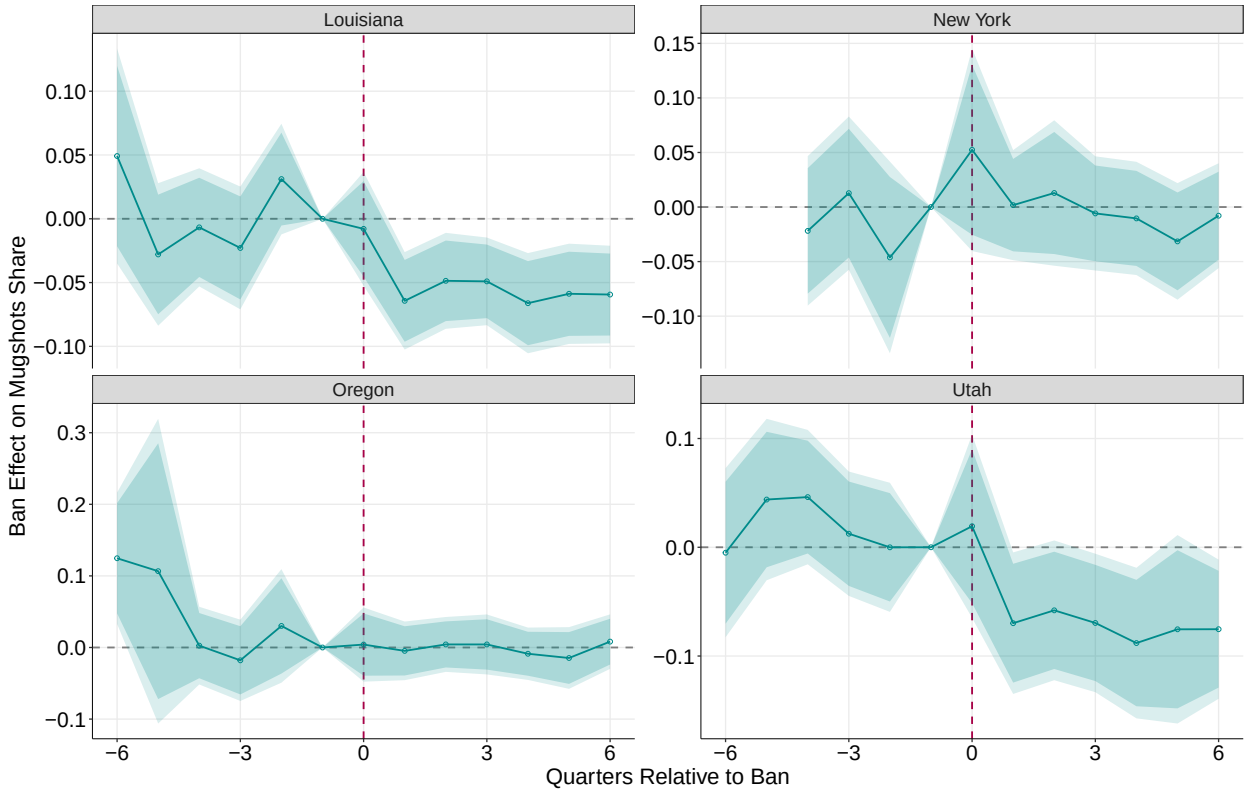
pooled two-way fixed effects regression would lead to the negative weighting problem highlighted by Goodman-Bacon (2021). We therefore estimate Columns 1, 4, and 5 using a stacked design. Across all crime articles, these columns show that mugshot bans reduce the share of mugshots published by 4 to 7 percentage points, a 30% to 40% decrease relative to the mean. Columns 2, 3, and 6 report estimates from a Callaway and Sant'Anna (2021) regression, which are still negative but smaller in magnitude and not statistically significant.

Given the coverage these policies mainly affect, the effect on all crime articles likely underestimates their true impact. Indeed, mugshot restrictions affect the release of booking photographs, which are taken when an individual is arrested and can circulate while the case is pending. Therefore, these policies should mainly affect coverage of the early stages of a case, and their effect should dilute in coverage of trials, sentencing, and appeals, where the accompanying images are typically not booking photographs at all. To verify this, we use the classification described in Section 2.1, which assigns every article to a stage of the criminal procedure and a crime type, and re-estimate the first stage separately within each category. Appendix Figure B.12 shows that, as predicted, the reduction is concentrated at the arrest and charging stage and is indistinguishable from zero at later stages. Further, the effect appears across crime types rather than being driven by any single category. We therefore report each subsequent specification for both all crime articles and the arrest-and-pretrial subsample, and treat the latter as the main sample.

We report the results from this subsample in Panel B of Table 3. The estimates are larger in magnitude than those in Panel A and remain stable and significant across all columns. In particular, the effect estimated on the state-level panel shows a decrease of more than 10 percentage points, corresponding to a 52% decrease relative to the baseline mean (Column 3). Similarly, the article-level panel shows a decrease of around 6 percentage points, corresponding to 27% of the baseline mean. Appendix Figure B.11 shows a similar pattern in the event-study regressions. Reassuringly, the figures show no pre-trend before the policy was approved in either sample. Additionally, the decrease is more pronounced only for the sample of articles on the arrest and pretrial stage.

Finally, we estimate the effect separately for each state to investigate any potential heterogeneity in policy adoption. Figure 6 shows substantial heterogeneity across states. In particular, while all states show the absence of any pretrend before the policy adoption, the effect for New York and Oregon is statistically indistinguishable from zero, suggesting that the effect is entirely driven by Louisiana and Utah. Several factors could explain this heterogeneity. First, the policies differ in how strictly agencies enforce them, generating heterogeneity in how easy it is for newsrooms to obtain

Figure 6: Event Study. Effect of Mugshot Bans on Mugshot Publication by State



Notes. Event study of the effect of mugshot bans on mugshot publication for the full sample, separately by state. All event studies include state and quarter fixed effects. The figure shows 95% and 90% confidence intervals.

Table 4: Effect of Mugshot Bans on Mugshot-to-Arrest Representativeness Ratio

	Relative Mugshot Coverage					
	Utah			Louisiana		
	Black	White	B − W	Black	White	B − W
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. All crime stages						
Mugshot ban × Post	−0.282***	−0.030	−0.252***	−0.043**	−0.019	−0.025
	(0.023)	(0.025)	(0.028)	(0.017)	(0.016)	(0.018)
Observations	3,162	3,166	3,162	3,160	3,164	3,160
Pre-ban control RR	0.226	0.179	0.047	0.207	0.157	0.051
Pre-ban treated RR	0.485	0.153	0.332	0.235	0.170	0.065
R^2	0.254	0.329	0.198	0.252	0.325	0.195
Panel B. Arrest and Pretrial						
Mugshot ban × Post	−0.383***	0.034	−0.418***	−0.081***	−0.025	−0.056*
	(0.034)	(0.094)	(0.095)	(0.025)	(0.025)	(0.028)
Observations	3,113	3,117	3,113	3,112	3,116	3,112
Pre-ban control RR	0.337	0.291	0.045	0.314	0.249	0.065
Pre-ban treated RR	0.580	0.184	0.396	0.361	0.250	0.110
R^2	0.212	0.179	0.151	0.209	0.178	0.151
State FE	X	X	X	X	X	X
Month FE	X	X	X	X	X	X
Article crime-type mix	X	X	X	X	X	X

Notes. Effect of mugshot bans on the mugshot-to-arrest ratio, separately by state and race. Columns 1 to 3 report the results for Utah, while Columns 4 to 6 for Louisiana. Columns 1 and 4 report the results for mugshots of Black arrestees, Columns 2 and 5 for those of White ones, and Columns 3 and 6 report the difference between the coefficients on Black and White mugshots. All columns include state and month fixed effects, and control for crime type. Panel A reports the results for all articles, while Panel B only for articles about crimes at the arrest or pre-trial stage. Standard errors are clustered at the state level (in parentheses): * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

booking photographs. Second, states also differ in the composition of the crimes they report, while mugshot publication concentrates on high-volume, low-severity offenses whose prevalence varies geographically (Section 3.2). Finally, these restrictions bind only on booking photographs released by agencies within the state, so an outlet with a strong preference for publishing mugshots can substitute coverage of arrests made elsewhere and leave its overall mugshot share unchanged. In the remainder of the analysis, we restrict attention to Utah and Louisiana and report them separately rather than pooled since both the composition of crimes and the racial composition of arrests differ substantially between the two states.

4.2.2 Mugshot Bans and Racial Disparities

Next, we ask whether restricting the release of mugshots changed how much mugshot coverage Black and White arrestees receive relative to their share of arrests. We estimate regression (2) separately for Utah and Louisiana, since both the crime mix and the racial composition of arrests differ substantially across the two states, and report the results in Table 4. In the full sample (Panel A), the ban reduces the Black mugshot-to-arrest RR in Utah by 0.282 while leaving the White RR essentially unchanged. This translates into a narrowing of the Black–White representation disparity by 0.252, corresponding to a 76% decrease in the state's overrepresentation of Black mugshots. In the arrest-and-pretrial sample (Panel B), the effect is larger, as in Table 3. The Black RR falls by 0.383 while the White ratio does not move, and the Black-to-White gap in RR narrows by 0.418, slightly more than Utah's entire pre-ban gap. Louisiana moves in the same direction, although the estimates are much smaller in magnitude. In particular, in Panel B, the Black RR decreases by 0.081, corresponding to 22% of the pre-ban average RR, with the Black-to-White gap narrowing by half.

To understand these magnitudes, it is useful to compare each treated state's pre-ban Black-to-White RR difference with that of never-treated states. Appendix Figure A.10 shows that the difference for both Utah and Louisiana is higher than the average of never-treated states before the bans were implemented. After the bans, both states converge to the national level of racial over-representation in mugshot coverage, but cannot reverse this over-representation.

Overall, the results show that mugshot coverage per arrest of Black individuals decreased more than that of White individuals after mugshot bans. This suggests that racial differences in mugshot reporting are not a fixed feature of crime coverage, but can shift following race-blind policies that affect organizations outside the newsroom. However, the scope for such a policy in affecting news reporting is bounded by initial state-level conditions. Utah started from a large disparity in racial coverage of mugshots and almost closed it after the ban was approved. Conversely, Louisiana displayed a lower level of Black over-representation in mugshot coverage before the ban, leading to a more muted effect of such policies.

Effect on Articles Narrative. To understand the effect of these policies on crime reporting more broadly, we move beyond mugshots and ask what coverage looked like once booking photographs were no longer available. We start by investigating what filled the space the mugshot used to occupy, distinguishing between non-mugshot images portraying individuals and images without

people. Next, we look at the composition of the articles in terms of the stage of the judicial process of the crime. Since booking photographs are produced at arrest, restrictions on them may push newsrooms to focus more on later stages, like trials, and shift away from early crime reporting.

To investigate this dimension, Appendix Figure B.16 reports the results from the reduced-form regression (2), including outlet fixed effects, and using the image and article composition as outcomes. Looking at image composition, outlets in both states publish more articles with images that do not portray people, with outlets in Louisiana publishing even fewer articles portraying people overall. Next, the results show a change also in the article composition. Consistent with the first-stage effect being concentrated at the arrest and charging stage, outlets publish fewer articles on arrests after the mugshot ban, and substitute these with more articles on investigations and, for Utah only, on the trial and outcome of the judicial process. Restricting the release of booking photographs therefore not only changes the images that are used in crime reporting, but also changes which stages of a criminal case are covered.

Finally, motivated by the descriptive analysis in Section 3.3, we investigate whether removing mugshots changes articles' content. For this purpose, we use the calibrated probability that an article's main image is a mugshot, predicted from the crime stage, the crime type, and the masked article text described in Section 2.3. Since this probability is computed from the article's own text and crime characteristics, it is defined for every crime article whether or not it carried an image, and therefore reflects changes in the article itself rather than in the image published alongside it.

We report the results from the reduced-form regression using this in Table 5. The predicted probability that an article's main image is a mugshot decreases in Utah by 6.7 percentage points, 32% of its baseline mean, and by 1.9 percentage points after including outlet fixed effects (Columns 1 and 2), with more muted effects for the arrest and pre-trial sample. Similarly, Panel A in Appendix Figure B.15 shows that the decline in predicted probability appears in the quarter following the policy approval and persists over time, while at the same time showing only a slight pre-trend going in the opposite direction of the main effect. At the same time, the results show no effect for Louisiana across all crime stages, consistent with the much weaker first stage documented in Section 4.2.1, and a positive one if we only focus on articles about arrests. The effects are consistent when we use only articles with images in Columns 3 and 6, restricting the sample to the same one used in the first stage.

This negative effect on the predicted probability of mugshots in Utah can be driven by two different channels: an article can look less like a mugshot story because it is written differently or

Table 5: Effect of Mugshot Bans on the Text-Predicted Probability of a Mugshot

	Utah			Louisiana		
	Full		Img. Only	Full		Img. Only
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. All crime stages						
Mugshot ban × Post	−0.067***	−0.019***	−0.014***	−0.009**	0.009***	0.003
	(0.004)	(0.002)	(0.003)	(0.004)	(0.002)	(0.002)
Observations	2,385,236	2,385,232	1,377,188	2,394,129	2,394,125	1,392,046
R^2	0.036	0.121	0.132	0.036	0.121	0.132
Pre-ban mean, State	0.2104	0.2104	0.2029	0.2009	0.2009	0.2079
Pre-ban mean, control	0.1474	0.1474	0.1671	0.1409	0.1409	0.1574
Panel B. Arrest and pretrial						
Mugshot ban × Post	−0.082***	−0.012***	−0.007*	−0.009	0.020***	0.010***
	(0.007)	(0.003)	(0.004)	(0.006)	(0.003)	(0.003)
Observations	873,815	873,810	522,236	878,213	878,208	529,049
R^2	0.074	0.219	0.235	0.076	0.219	0.235
Pre-ban mean, State	0.3254	0.3254	0.3085	0.3275	0.3275	0.3373
Pre-ban mean, control	0.2333	0.2333	0.2662	0.2295	0.2295	0.2582
State FE	X	X	X	X	X	X
Month FE	X	X	X	X	X	X
Outlet FE		X	X		X	X

Notes. Effect of mugshot bans on the predicted probability that an article's main image is a mugshot. The outcome is the calibrated probability produced by the combined model of Section 2.3, which predicts mugshot publication from the crime stage, the crime type, and the masked article text, and is therefore measured from the written article alone. Columns 1 to 3 report the results for Utah and Columns 4 to 6 for Louisiana. All columns include state and month fixed effects, and Columns 2, 3, 5, and 6 also include outlet fixed effects. Columns 3 and 6 restrict the sample to articles that carry an image, matching the denominator of the first stage. Panel A reports the results for all articles, while Panel B only for articles about crimes at the arrest or pre-trial stage. Pre-ban means are computed over the pre-policy period of each estimation sample, separately for the treated state and for the never-treated states. Standard errors are clustered at the state level (in parentheses): * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

because it covers a different set of cases. To disentangle these channels, we use the decomposition of the overall predicted probability into the *incident* component, which moves only when the mix of cases changes, and the *text narrative* component, which captures what the wording adds while holding case characteristics fixed. Appendix Table B.4 reports the results and shows that, in Utah, the effect on the joint probability is driven equally by both components. Therefore, Utah's newsrooms reported on the same types of cases differently after the ban, in a way that reads less like articles that historically carried a booking photograph, which, as Section 3.3 shows, is the more perpetrator-centered and person-named form. Conversely, the joint positive coefficient in Louisiana is entirely driven by the incident component, while the text component is small and indistinguishable from zero, suggesting that removing access to booking photographs may have led outlets to move toward reporting on more mugshot-prone crime types.

4.2.3 Mugshot Bans and Crime Perceptions

The evidence so far shows that mugshot bans decreased the use of such images in crime articles, changed the racial representation in these articles, and, in Utah, even affected the narrative. Recent literature shows that visual content is processed faster than text and carries persuasive power of its own (Thorpe et al., 1996; Caprini, 2024; Ciccarone, 2025), and images of criminal suspects in particular have been shown to activate the association between race and crime (Eberhardt et al., 2004). Therefore, in the final part of the paper, we investigate whether the reduction in mugshot publication translates into a change in individuals' perceptions about crime.

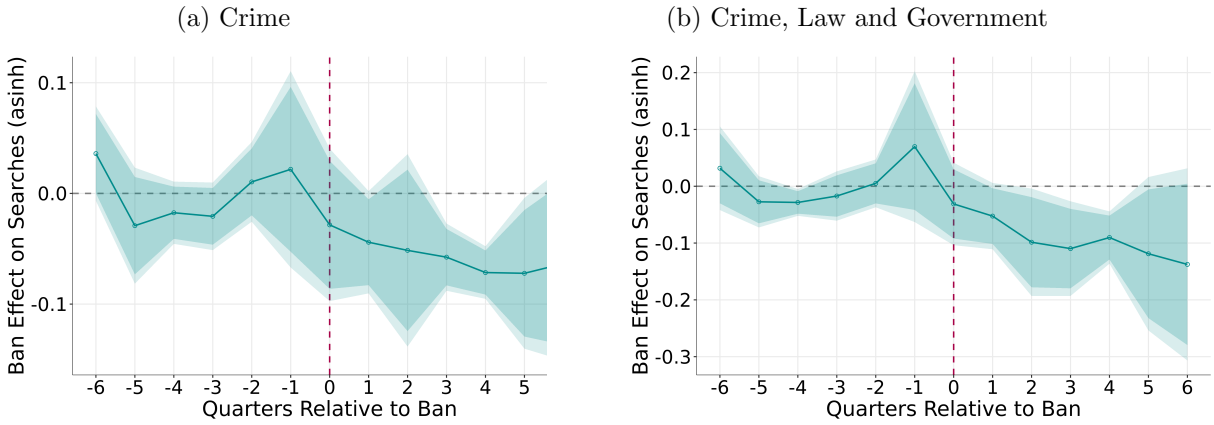
For this purpose, we use Google Trends, which measures how intensively a given query is searched in a given state and month, and we take the search intensity for crime-related queries as a measure of how salient crime is to the residents of a state.[9] We use two queries: "crime", and the same query restricted to Google's Law and Government category.[10] Using these as outcomes, we estimate the

[9]Google Trends reports the share of searches for a query within a region and period, normalized to a 0–100 scale over the requested window and rounded to integers, so series obtained from different requests are not directly comparable and infrequent queries can be rounded down to zero. We therefore construct the panel in three steps, following recommendations from Choi and Varian (2012), Stephens-Davidowitz (2014), and Hölzl, Keusch and Sajons (2025). First, we collect levels across states from a single regional request, which returns all states on one scale. Then, we run state-specific requests to create a within-state monthly series. Finally, we collect search frequencies for our main queries relative to a list of unrelated and popular anchor terms, following the anchor-bank procedure of West (2020). Our panel covers the fifty states and the District of Columbia at monthly frequency from 2016 to 2025. We assess the reliability of our queries by repeating the pull on separate days and obtain a correlation of 0.990 across queries. Since the index is zero in a non-negligible share of state-quarter cells, we take the inverse hyperbolic sine of the index, so that coefficients approximate proportional changes.

[10]This restriction excludes searches about crime related to entertainment. At the same time, the category assignment is not validated, but its correlation with the unrestricted series is 0.81. The restriction therefore removes a minority share of searches and leaves the time variation we exploit largely intact.

same staggered difference-in-differences design as above at the state-quarter level.

Figure 7: Event Study. Effect of Mugshot Bans on Crime-Related Searches



Notes. Event study of the effect of mugshot bans on the intensity of Google searches for crime-related queries, pooling the four treated states and using never-treated states as controls. The outcome is the inverse hyperbolic sine of the state-quarter search index. Estimates are obtained with the Callaway and Sant'Anna (2021) estimator and aggregated by quarter relative to the ban. Panels A and B use the query "crime" and the same query restricted to Google's Law and Government category, respectively. The figure shows 95% and 90% confidence intervals based on a multiplier bootstrap with standard errors clustered at the state level.

Figure 7 reports the results. Searches for crime show no systematic pre-trend before the ban, decline steadily in the quarters after the policy is approved, and stabilize thereafter. The pooled effect is −0.073 (0.018), which for this transformation corresponds to a decline of about 7 percent in crime-related search intensity. We find the same pattern once the query is restricted to the Law and Government category, where the estimate is larger, −0.102 (0.035), though less precisely estimated and preceded by an uptick in the quarter immediately before the ban.[11]

A main concern is that these results may capture other changes occurring in the treated states at the same time. To address this, we estimate the same specification on a set of placebo queries unrelated to crime: unemployment, health insurance, inflation, and war. Appendix Figure B.18 shows that none of these changes are significant after the policy approval date, and that their point estimates are inconsistent in sign, alleviating concerns about the effect being driven by a more general decrease in search volumes or by co-occurring events. The last of these is a demanding test, since the post-treatment window contains the beginning of the war in Ukraine, which the design nonetheless differences out. Conversely, searches for queries that capture attention to the criminal justice system, like "arrest" and "police", decline alongside the crime queries (Appendix

[11]Appendix Figure B.17 reports the same event study estimated separately for each treated state, and shows that Utah displays the cleanest effect.

Figure B.19).

Finally, we provide survey evidence consistent with these results. For this purpose, we use the probability that a respondent names crime as the most important problem facing the country from the Gallup Social Series. Appendix Figure B.20 shows that this probability falls in Utah after the ban and is unchanged in Louisiana, while a set of placebo survey outcomes shows no effect (Appendix Figures B.22 to B.24).[12]

Overall, these results suggest that restricting the publication of mugshots reduces the salience of crime among residents of the treated states, and that this reduction is specific to crime rather than reflecting a general change in what residents search for. However, search behavior is only a proxy for perceptions, and the survey data are underpowered to provide robust evidence on this, so we treat this evidence as preliminary.

5 Conclusion

As news consumption becomes increasingly visual, the images that accompany crime reporting can help shape how readers perceive crime and those accused of it. Mugshots are among the most distinctive of these images. These standardized booking photographs, which newsrooms obtain from law enforcement and republish at their discretion, often reflect editorial choices rather than arrests alone. Understanding how these choices are made, and whether they lead to disparities in representation across racial groups, is a first-order question for a media environment where images are becoming prevalent.

This paper provides large-scale evidence on this question. Combining 3.46 million crime articles with a face-level pipeline for mugshot detection, race inference, and individual clustering, we document four sets of findings. First, the share of crime articles carrying mugshots has declined over our sample period. Second, relative to their share of arrests in NIBRS, Black arrestees are over-represented in mugshot coverage compared to White arrestees, with the gap concentrated in high-volume, low-severity offenses such as DUI and drug crimes, namely those crimes for which publication is most discretionary. Third, exploiting staggered state-level restrictions on mugshot disclosure, we find that these policies reduce mugshot publication by about a third and narrow the over-representation of Black arrestees, with an effect ranging between half and three quarters, depending on the state. Fourth, crime coverage adjusts along other margins: outlets substitute

[12] A similar pattern holds under a broader definition that also counts drugs, the judicial system, police brutality, and school shootings as crime, though the estimates are noisier (Appendix Figure B.21).

other imagery for booking photographs and report fewer arrests, and in Utah the language of crime articles shifts along with the pictures. Finally, we provide suggestive evidence from Google searches of a decrease in the salience of crime in the treated states.

Whether these changes in coverage reach readers remains open. Search behavior is only a proxy for what residents believe, and establishing whether the disparities we document carry downstream consequences for public opinion and judicial outcomes requires more fine-grained data. For policy, our findings support the view that restricting mugshot disclosure can reduce both the volume and the racial skew of booking-photo coverage, a question courts and newsrooms have so far debated with little systematic evidence. At the same time, the effects concentrate in the states where publication actually fell and where Black over-representation was largest, due to potential differences in policy enforcement and crime news, so a one-size-fits-all restriction will not mechanically narrow representation gaps everywhere.

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A Descriptives

Table A.1: Validation performance of the text and image classifiers.

Task	**Model + features**	**Headline metrics**
Text classifiers		
Stage of criminal procedure	Eight-class multinomial logistic regression, TF-IDF; predictions aggregated to four stages	human-validation accuracy 0.721; macro-F1 0.718; weighted-F1 0.726; Cohen's κ 0.620
Crime type (20 classes, multi-label)	One-vs-rest logistic regression, TF-IDF	human-validation micro-F1 0.613; macro-F1 0.557; exact-set accuracy 0.361; primary-type-in-human-set accuracy 0.729
Image classifiers		
Mugshot detection (binary, face-level)	EfficientNet-B0 fine-tuned on manually validated face crops; production threshold $\tau = 0.90$	accuracy 0.920; precision 0.984; recall 0.848; F1 0.911
Race / ethnicity (6 classes, face-level)	3-layer MLP head on FaceNet 512-d embeddings, trained on FairFace	aggregate precision 0.731; per-class precision / recall: White 0.701 / 0.746; Black 0.852 / 0.844; Asian 0.891 / 0.909; Latino-Hispanic 0.490 / 0.493; Indian 0.701 / 0.669; Middle Eastern 0.576 / 0.521

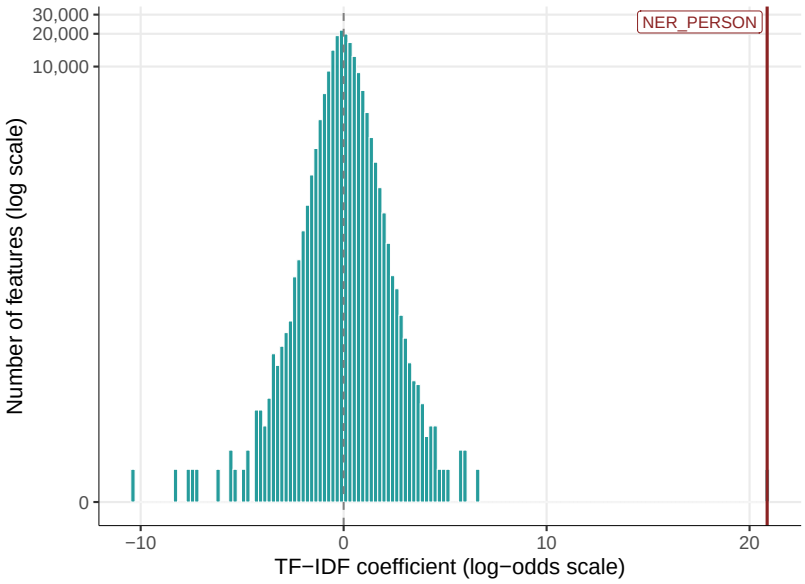
Text-classifier metrics compare production predictions with pooled human judgments from the stratified 497-article validation sample. The stage metrics aggregate investigation; arrest or charging; court proceedings; and post-case proceedings. The crime-type metrics retain the production twenty-label taxonomy. Image-classifier metrics are computed on manually validated face crops or, for the face-attribute heads, on the FairFace evaluation set. All threshold-dependent mugshot metrics use the production threshold of 0.90.

Table A.2: Performance of the Narrative-Focus Sentence Classifier

	Held-out GPT labels				Human annotations			
Class	Precision	Recall	F1	Support	Precision	Recall	F1	Support
Other	0.722	0.726	0.724	175	0.923	0.889	0.906	27
Victim-centered	0.845	0.750	0.795	116	0.867	0.812	0.839	16
Perpetrator-centered	0.848	0.865	0.856	386	0.938	0.918	0.928	49
Mixed	0.533	0.571	0.552	56	0.200	0.500	0.286	2
Accuracy	–	–	0.791	733	–	–	0.883	94
Macro average	0.737	0.728	0.732	733	0.732	0.780	0.739	94

Notes. The classifier is the production two-member DeBERTa-v3-base ensemble trained on GPT-5.6 Luna labels, with independently initialized members (seeds 20260822 and 20260823). The held-out evaluation uses 733 story-disjoint sentences that were not used for model training or threshold selection. The human evaluation is a blinded random audit of 100 corpus predictions; six annotations marked unclear are treated as abstentions, leaving 94 evaluable sentences. No sentence or article in the human audit overlaps the GPT-labeled development corpus. Predictions use the validation-selected thresholds of 0.75 for victim-centered and 0.65 for perpetrator-centered language. The held-out GPT-label results measure agreement with the teacher model, while the human results provide an external construct-validity check. The human-validation sample contains only two mixed sentences, so performance for this class should be interpreted cautiously.

Figure A.1: Masked-Text Coefficients in the Mugshot-Prediction Model

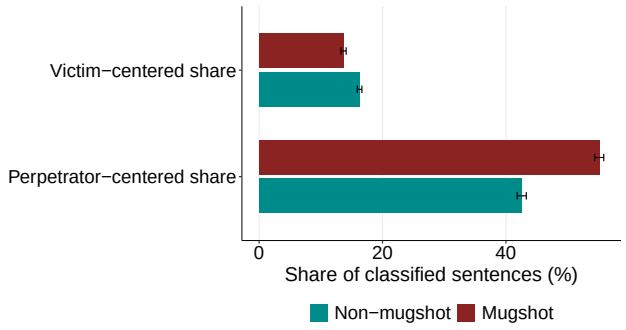


Notes. Distribution of the regularized TF–IDF log-odds coefficients in the combined model's text component. The brown line marks `NER_PERSON`; the dashed line marks zero.

Table A.3: Time of Mugshot Policies

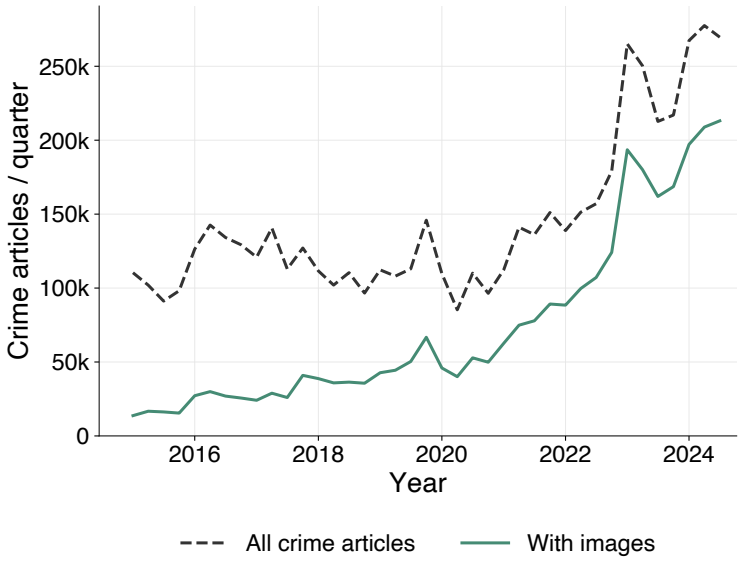
State	Approval	Implementation	Reversal
New York	March 2019	March 2019	-
Utah	February 2021	July 2021	-
Oregon	July 2021	January 2022	-
Louisiana	June 2022	August 2022	August 2024

Figure A.2: Raw Narrative Focus by Mugshot Status



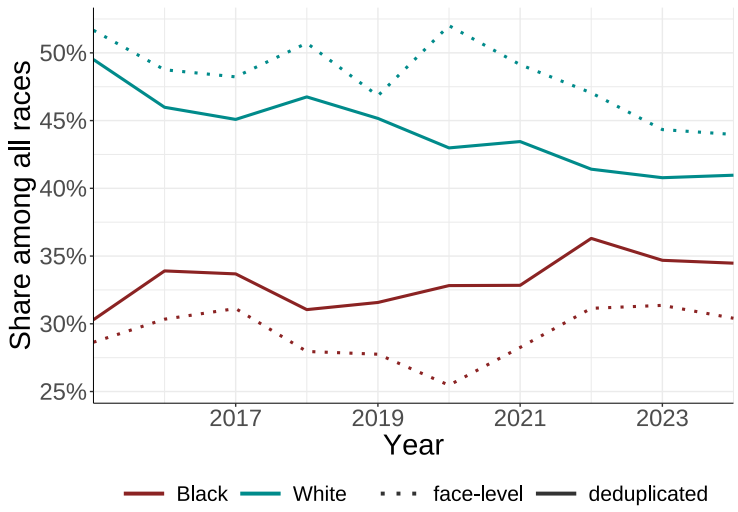
Notes. Article-weighted shares of victim- and perpetrator-centered sentences in the cleaned main analysis sample, predicted by the two-member DeBERTa-v3-base ensemble. Whiskers show outlet-clustered 95% confidence intervals.

Figure A.3: Crime articles and articles with images



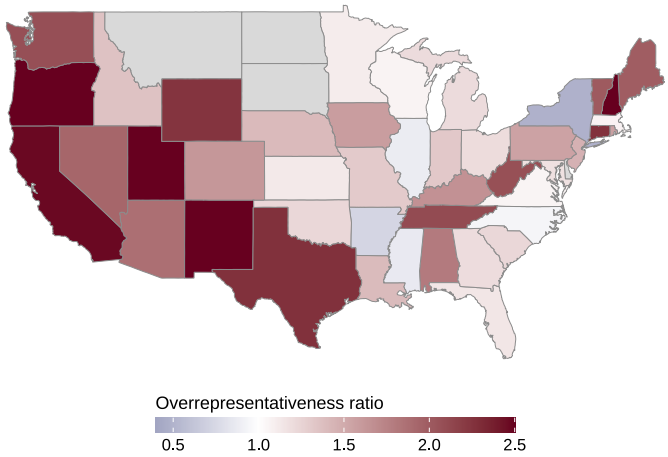
Notes. Number of articles filtered for crime incidents between 2015 and 2024 as well as the number of images collected from articles per quarter.

Figure A.4: Share of Black versus White individuals in mugshots at the impression and individual levels.



Notes. The lines show, separately for faces classified as Black (red) and White (green), each group's share among all predicted-race categories in published mugshots. Dotted lines count every mugshot-face impression, including repeated publication of the same person; solid lines deduplicate impressions to inferred individuals using the face-clustering procedure described in Section 2.2. Because other predicted-race categories remain in the denominator, the Black and White shares need not sum to 100%.

Figure A.5: State-level representation of Black-vs-White mugshots relative to arrests



Notes. Ratio of the Black mugshot-to-arrest ratio to the White mugshot-to-arrest ratio, (i.e., the ratio of ratios: $(black_mugshots/black_arrests)/(white_mugshots/white_arrests)$) by state between 2022 and 2024. The data is restricted to outlets with a local audience, and the state associated with each mugshot is determined by the outlet headquarters location. State-level arrest counts are determined by the location of the reporting police station. States with fewer than 100 mugshot faces are shown in grey.

Figure A.6: Total NIBRS arrests by crime type and race

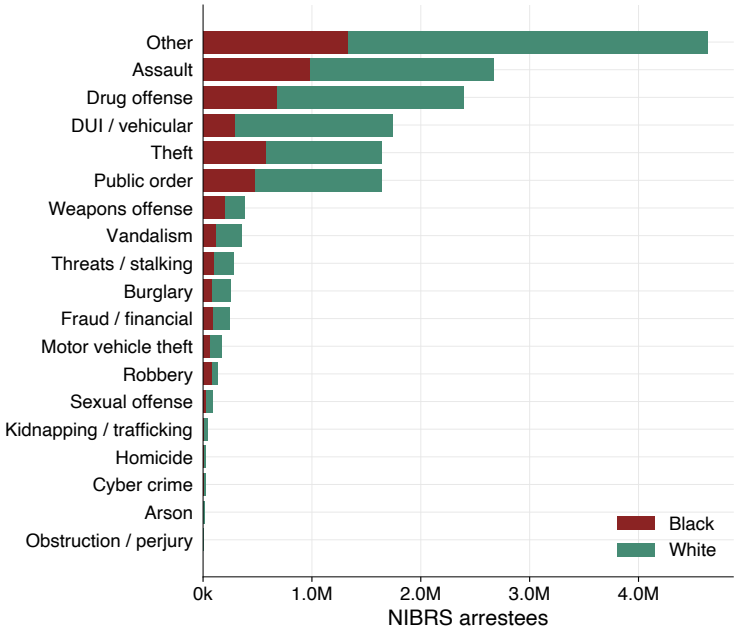
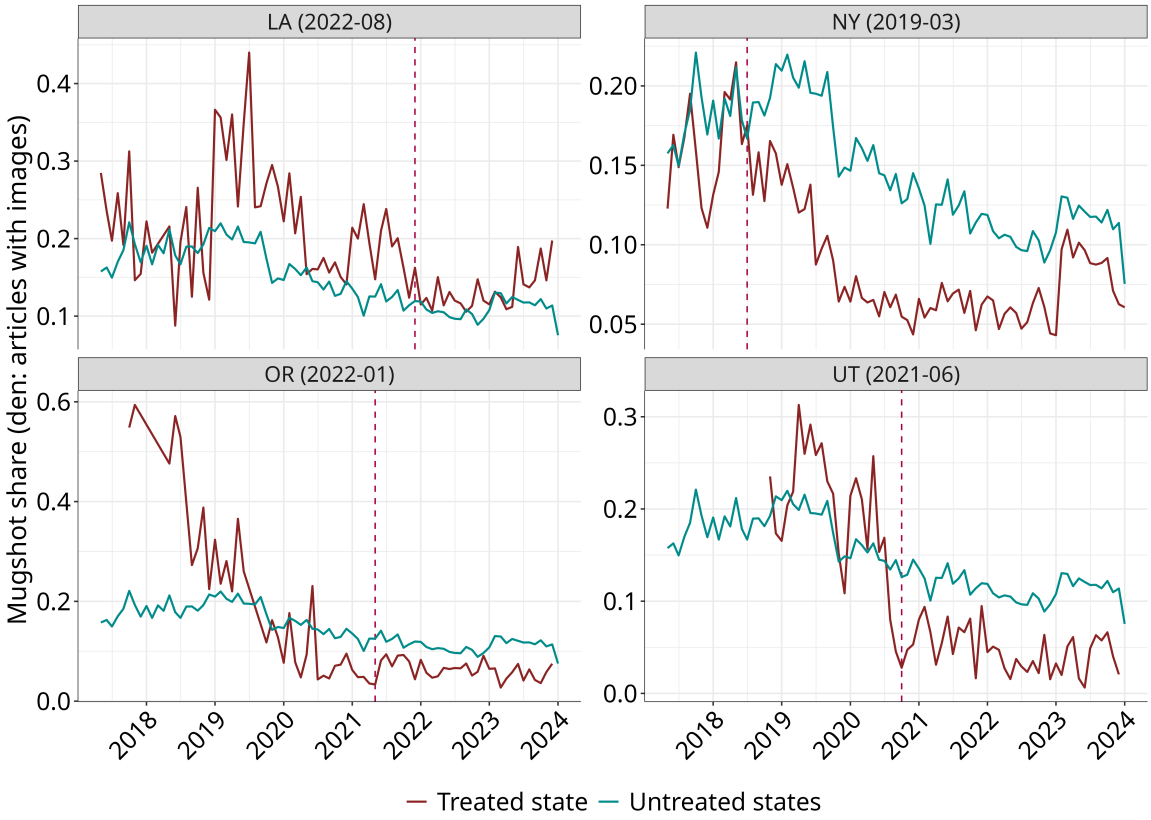


Figure A.7: Share of Mugshots over Time by Treated States

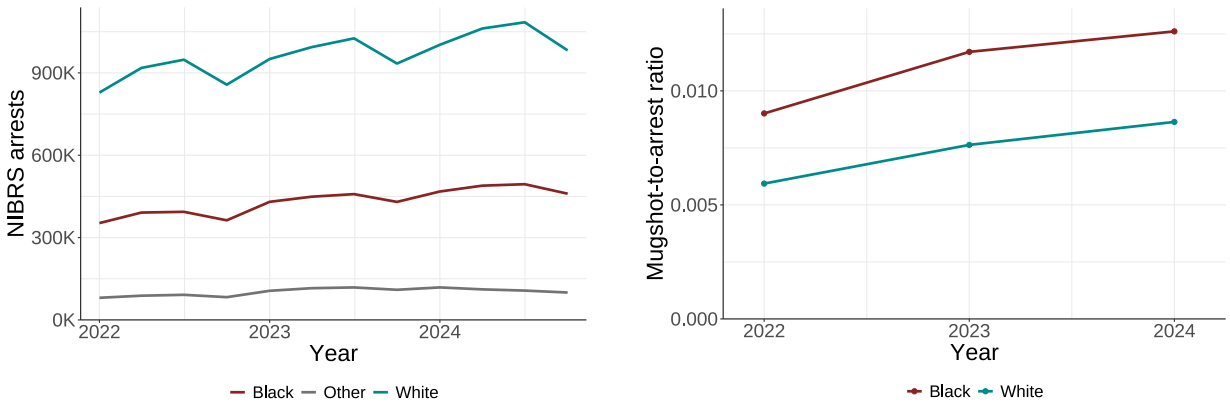


Notes. Share of mugshots over time plotted separately by treated and untreated states. Each panel plots the trend for the different treated states. The dashed gray lines indicate the date of approval of the policy, while the dashed red line indicates the reversal.

Figure A.8: Quarterly NIBRS arrests and article-level mugshot-to-arrest ratios by race

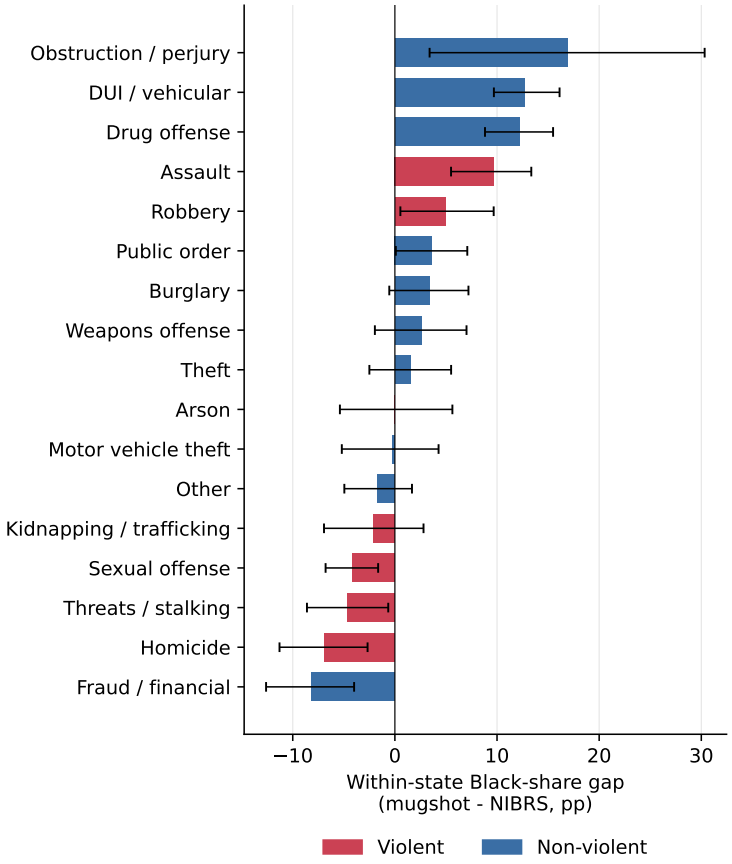
(a) NIBRS arrestees by race

(b) Mugshot-to-arrest ratio by race



Notes. Panel A.8a shows quarterly NIBRS arrest records by race from 2022 through 2024. Panel A.8b divides, separately by race and year, the number of crime articles displaying at least one mugshot face classified in that racial category by the corresponding number of NIBRS arrests. It is therefore an article-level publication-to-arrest ratio, not a count of unique people or mugshot-face impressions.

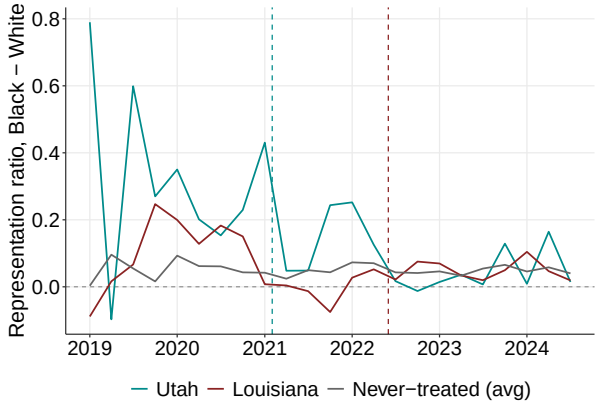
Figure A.9: Black vs White representation in mugshots by crime type.



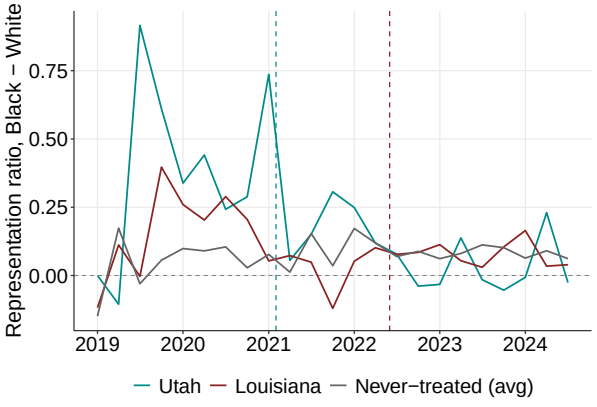
Notes. For each state and crime type, the outcome is the Black share among Black and White mugshot-face impressions in local-outlet articles minus the corresponding Black share among Black and White NIBRS arrests, in percentage points. Each detected mugshot face in an article is one impression; repeated appearances are retained, and multiple mugshot faces in one article contribute separately. Articles enter every qualifying predicted crime label, so an impression can enter more than one crime-type comparison. The figure reports the unweighted mean across eligible states for 2022–2024, with state-bootstrap 95% confidence intervals. State–crime cells require at least 20 observations on both sides, and crime types require at least five states.

Figure A.10: Black–White Representation Gap over Time

(a) All crime stages



(b) Arrest and pretrial



Notes. Quarterly difference between the Black and White representation ratios defined in Section 4.1 ($R^B_{st} - R^W_{st}$), for Utah, Louisiana, and the average across never-treated states. Vertical dashed lines mark the approval dates of the Utah and Louisiana policies, in the color of the corresponding series. The horizontal dashed line marks equal representation of Black and White arrestees relative to their arrest shares. New York and Oregon are excluded from the plot.

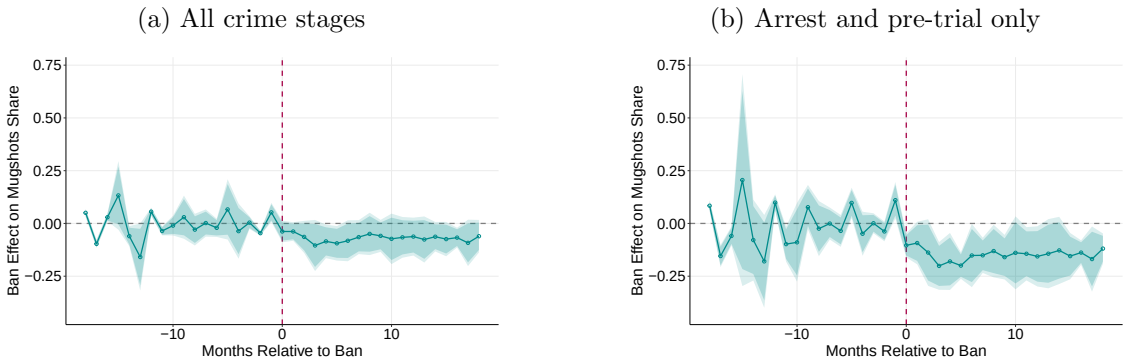
B Additional Results

Table B.4: Decomposing the Text-Predicted Probability of a Mugshot

	Utah			Louisiana		
	Incident	Text	Combined	Incident	Text	Combined
	(1)	(2)	(3)	(4)	(5)	(6)
All crime stages						
Mugshot ban × Post	−0.009***	−0.010***	−0.019***	0.007***	0.002	0.009***
	(0.001)	(0.002)	(0.002)	(0.001)	(0.001)	(0.002)
Observations	2,385,232	2,385,232	2,385,232	2,394,125	2,394,125	2,394,125
R^2	0.045	0.108	0.121	0.046	0.108	0.121
Pre-ban mean, treated state	0.1911	0.0192	0.2104	0.1748	0.0261	0.2009
Pre-ban mean, control	0.1646	-0.0171	0.1474	0.1610	-0.0201	0.1409
Arrest and pretrial						
Mugshot ban × Post	−0.005***	−0.007**	−0.012***	0.014***	0.006**	0.020***
	(0.001)	(0.003)	(0.003)	(0.001)	(0.003)	(0.003)
Observations	873,810	873,810	873,810	878,208	878,208	878,208
R^2	0.080	0.189	0.219	0.080	0.189	0.219
Pre-ban mean, treated state	0.2877	0.0376	0.3254	0.2764	0.0511	0.3275
Pre-ban mean, control	0.2573	-0.0240	0.2333	0.2570	-0.0275	0.2295
State FE	X	X	X	X	X	X
Month FE	X	X	X	X	X	X
Outlet FE	X	X	X	X	X	X

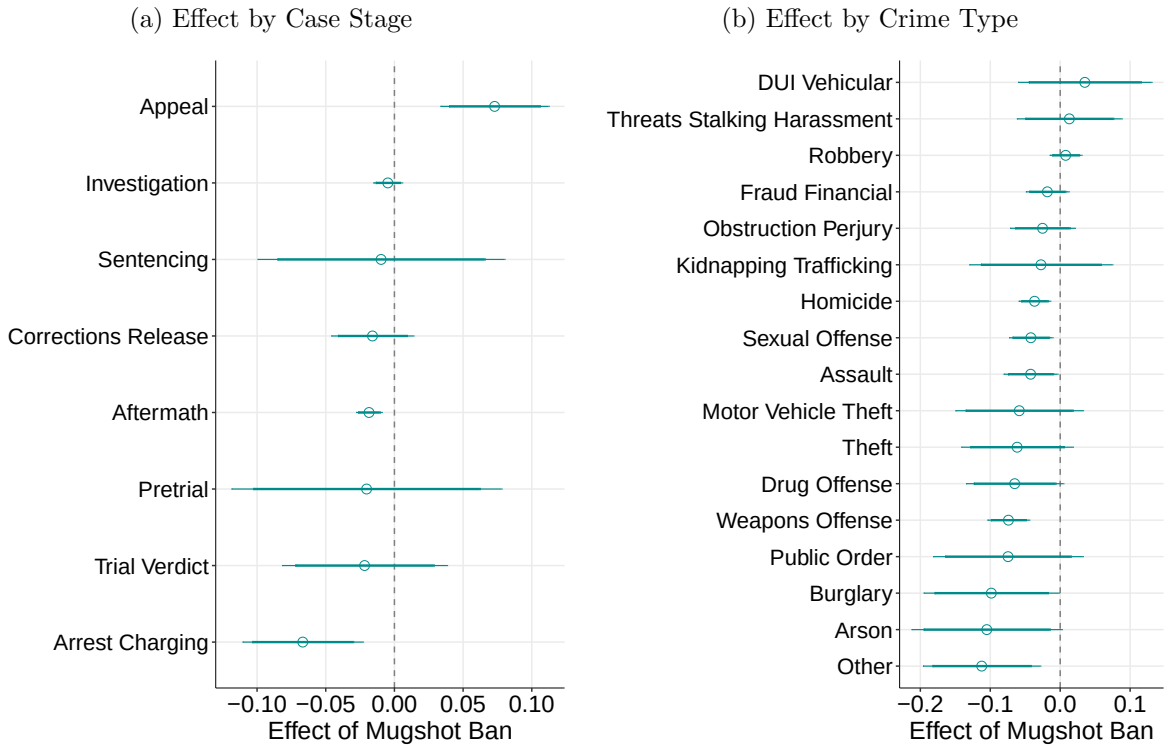
Notes. Effect of mugshot bans on the predicted probability that an article's main image is a mugshot, separately for its two components. The incident component is predicted from the crime stage and crime type only, the text component is the difference between the combined prediction and the incident one, and the two sum to the combined score. Columns 1 to 3 report the incident, text, and combined score, while Columns 4 to 6 repeat them adding outlet fixed effects. All columns include state and month fixed effects and use the full sample of crime articles. Panels report Utah and Louisiana separately, each for all articles and for articles about crimes at the arrest or pre-trial stage. Pre-ban means are computed over the pre-policy period, separately for the treated state and for the never-treated states. The last row of each panel reports wild cluster bootstrap p-values imposing the null. Standard errors are clustered at the state level (in parentheses): * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Figure B.11: Effect of Mugshot Bans on Share of Mugshots: Callaway and Sant'Anna (2021)



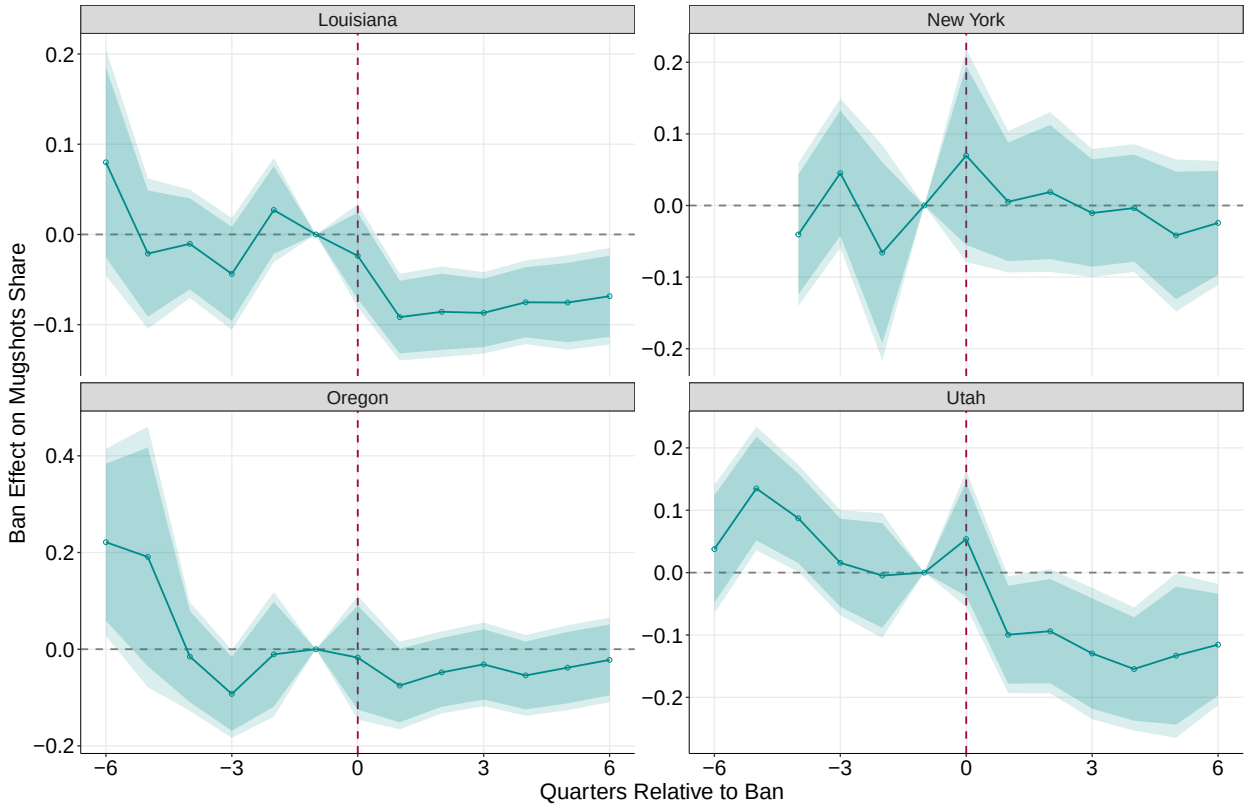
Notes. Event study of the effect of mugshot bans on mugshot publication. Event studies are estimated with the Callaway and Sant'Anna (2021) estimator. The unit of observation is the state–month and the outcome is the share of local crime articles with images carrying a mugshot. Panel A plots the dynamic ATT for the full sample of crime articles, while Panel B restricts the sample to articles at the arrest and pre-trial stages. Standard errors are clustered at the state level, using the multiplier bootstrap of Callaway and Sant'Anna (2021) with 1,000 iterations. The figure shows 95% and 90% confidence intervals.

Figure B.12: Effect of Mugshot Bans on Mugshot Publication, by Case Stage and Crime Type



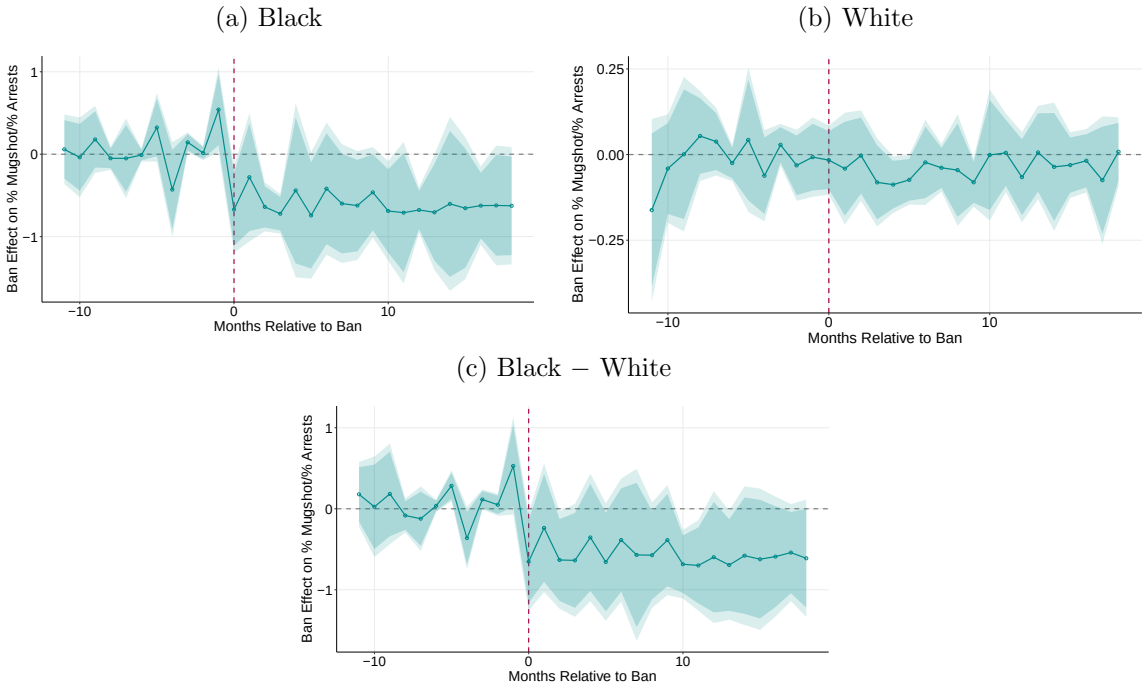
Notes. Effect of mugshot bans on mugshot publication separately by stage of the judicial process (Panel A) and crime (Panel B). All coefficients are estimated using a two-way fixed effects regression with month-to-event and state fixed effects. The figures show 90% and 95% confidence intervals constructed with standard errors clustered at the state level.

Figure B.13: Event Study. Effect of Mugshot Bans on Mugshot Publication by State—Arrest–Pre-Trial



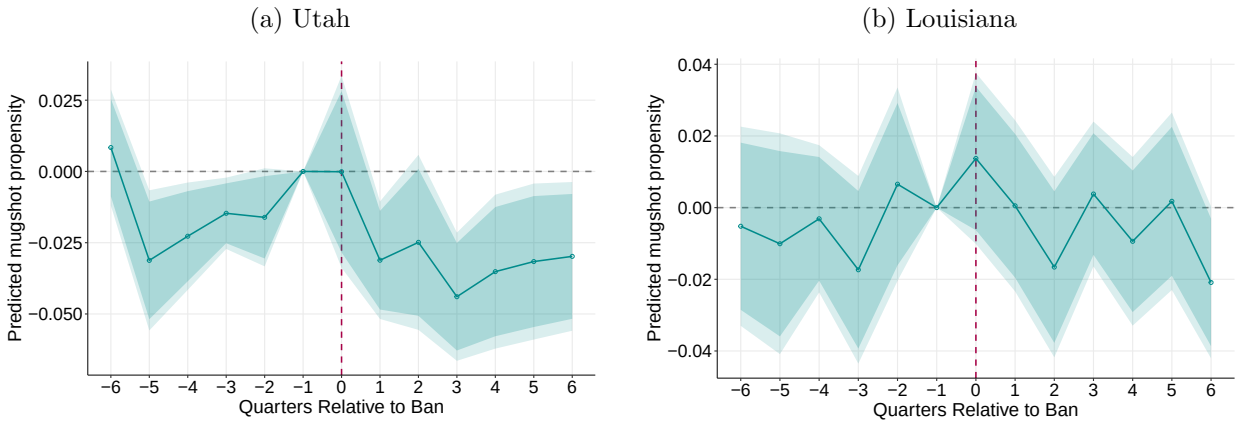
Notes. Event study of the effect of mugshot bans on mugshot publication for the full sample, separately by state and only for articles about crimes at the arrest or pre-trial stage. All event studies include state and quarter fixed effects. The figure shows 95% and 90% confidence intervals.

Figure B.14: Effect of Mugshot Bans on Racial Disparity in Mugshot Publication: Event Study, Callaway and Sant'Anna (2021)



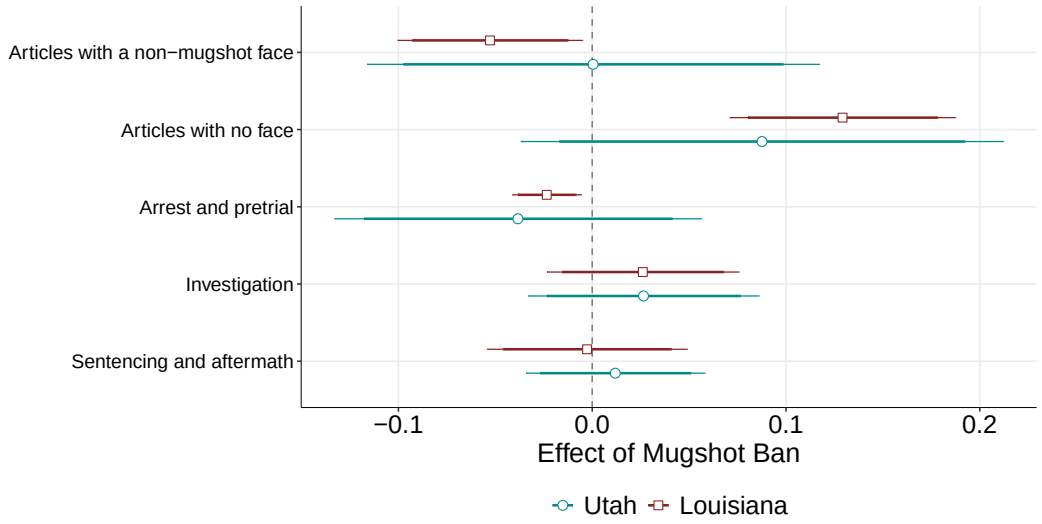
Notes. Event study of the effect of mugshot bans on the ratio between the share of mugshots and the share of arrests separately by race, for articles about crimes at the arrest or pre-trial stage. All event studies are estimated with the Callaway and Sant'Anna (2021) estimator using state and quarter fixed effects. Panels A and B plot the results for mugshots and arrests of Black and White individuals, respectively, and Panel C plots the difference between the two coefficients. The figure shows 95% and 90% confidence intervals.

Figure B.15: Event Study. Effect of Mugshot Bans on the Latent Mugshot-Story Propensity



Notes. Event study of the effect of mugshot bans on the predicted probability that an article's main image is a mugshot, for articles that carry an image. Event studies are estimated on outlet-by-quarter cells weighted by the number of articles, using outlet, state, and quarter fixed effects. Panels A and B plot the results for Utah and Louisiana, respectively. The figure shows 95% and 90% confidence intervals.

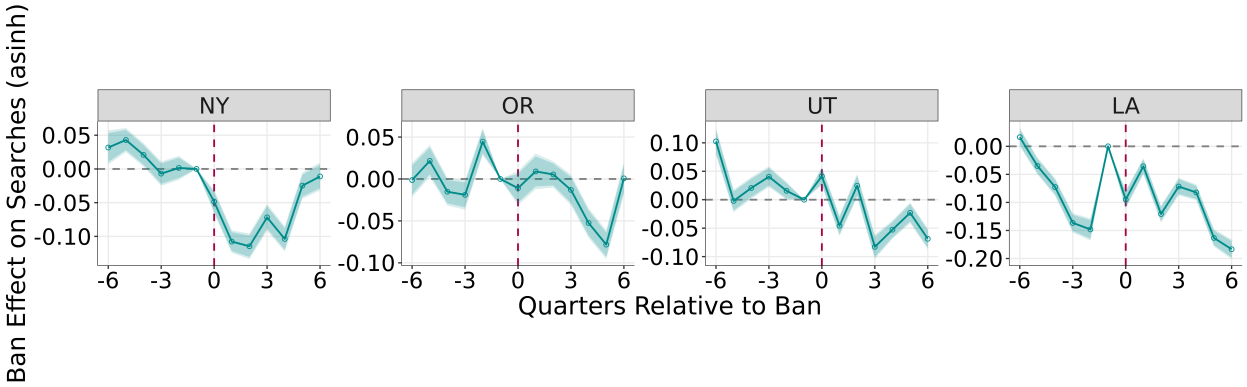
Figure B.16: Effect of Mugshot Bans on Image Composition and Case-Stage Coverage



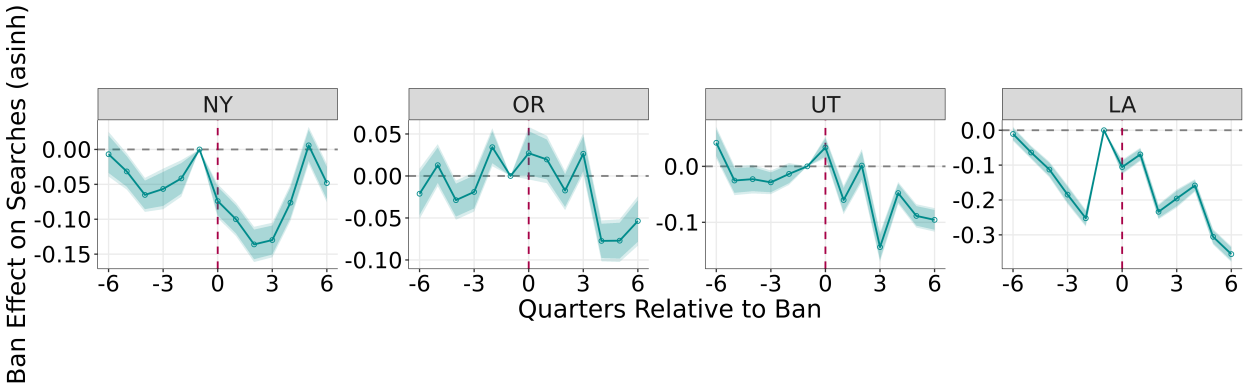
Notes. Effect of mugshot bans on the composition of crime coverage, estimated separately for Utah and Louisiana. Each coefficient is estimated including state, month, and outlet fixed effects, controlling for the log number of active outlets interacted with month. Bars are 90% and 95% confidence intervals with standard errors clustered at the outlet level.

Figure B.17: Event Study. Effect of Mugshot Bans on Crime-Related Searches, by State

(a) Crime

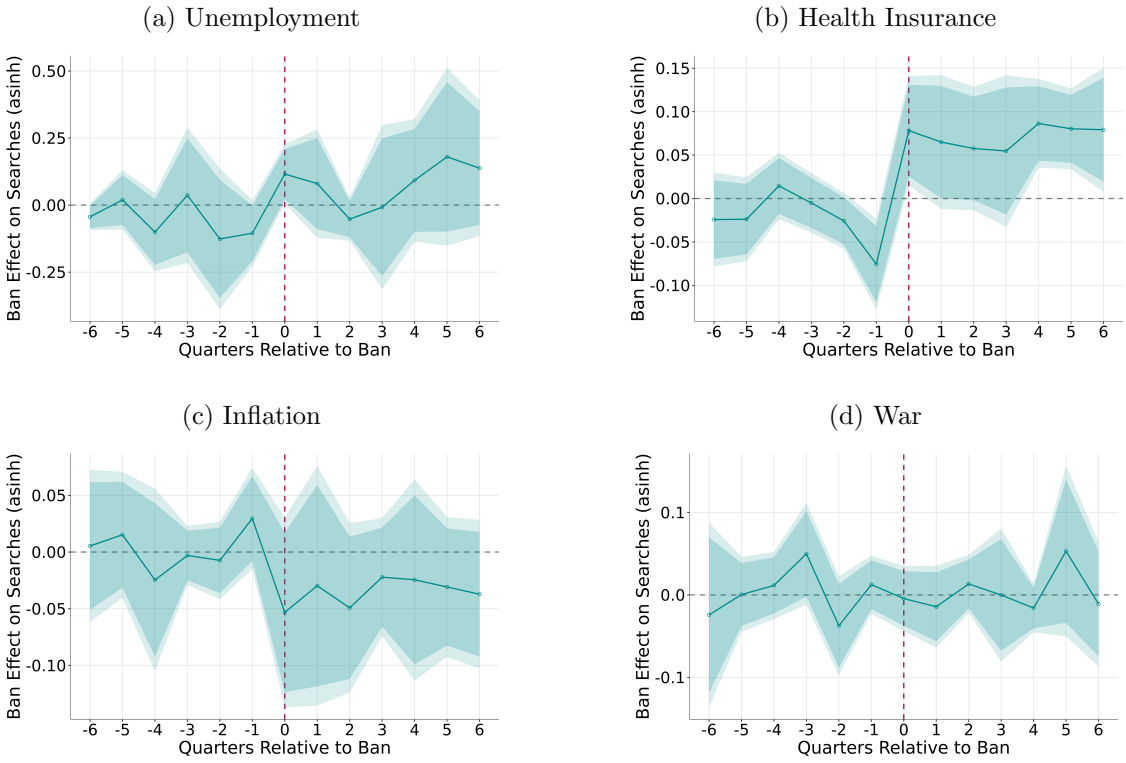


(b) Crime, Law and Government



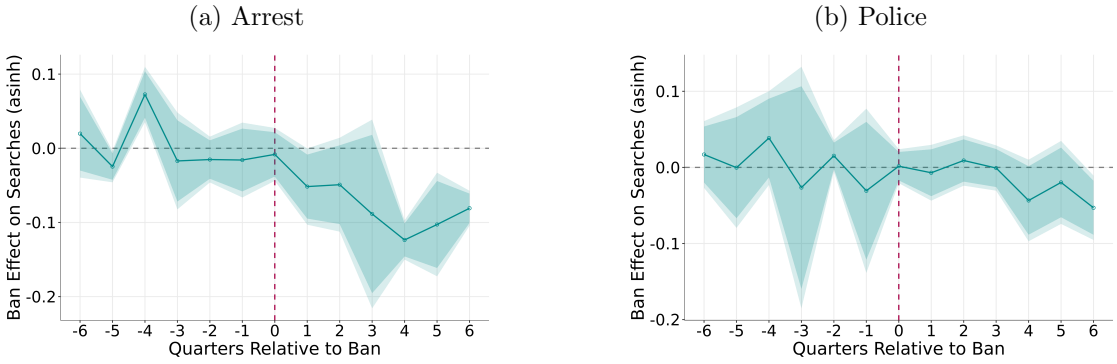
Notes. Event study of the effect of mugshot bans on the intensity of Google searches for crime-related queries, estimated separately for each treated state against the never-treated states. The outcome is the inverse hyperbolic sine of the state-quarter search index. All specifications include state and quarter fixed effects, and the quarter before the ban is the omitted category. Panels A and B use the query "crime" and the same query restricted to Google's Law and Government category, respectively. The four states shown are those adopting a statewide restriction, and are the ones entering the pooled estimates in Figure 7. Louisiana is truncated at the reversal of its restriction in August 2024. The figure shows 95% and 90% confidence intervals based on heteroskedasticity-robust standard errors.

Figure B.18: Event Study. Effect of Mugshot Bans on Placebo Searches



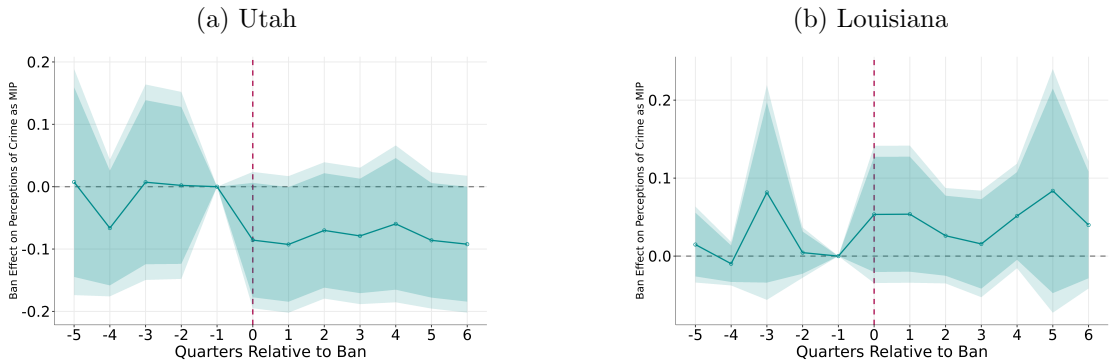
Notes. Event study of the effect of mugshot bans on the intensity of Google searches for four queries unrelated to crime, pooling the four treated states and using never-treated states as controls. The outcome is the inverse hyperbolic sine of the state-quarter search index. Estimates are obtained with the Callaway and Sant'Anna (2021) estimator and aggregated by quarter relative to the ban. Panels A, B, C, and D use the queries "unemployment", "health insurance", "inflation", and "war", respectively. The figure shows 95% and 90% confidence intervals based on a multiplier bootstrap with standard errors clustered at the state level.

Figure B.19: Event Study. Effect of Mugshot Bans on Criminal Justice Searches



Notes. Event study of the effect of mugshot bans on the intensity of Google searches for queries capturing attention to the criminal justice system, pooling the four treated states and using never-treated states as controls. The outcome is the inverse hyperbolic sine of the state-quarter search index. Estimates are obtained with the Callaway and Sant'Anna (2021) estimator and aggregated by quarter relative to the ban. Panels A and B use the queries "arrest" and "police", respectively. The figure shows 95% and 90% confidence intervals based on a multiplier bootstrap with standard errors clustered at the state level.

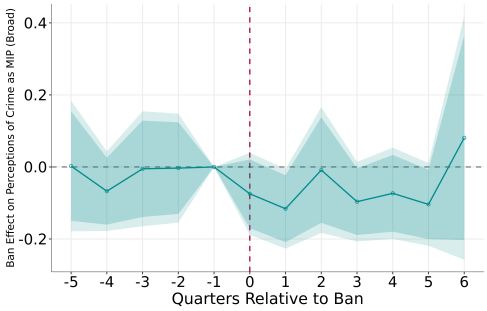
Figure B.20: Event Study. Effect of Mugshot Bans on Crime as the Most Important Problem



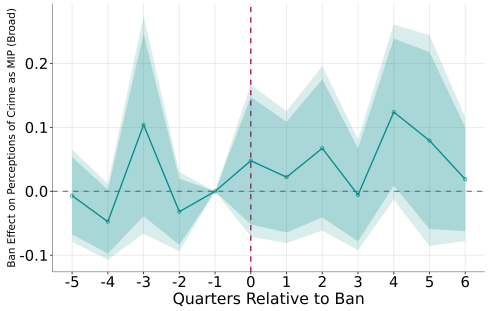
Notes. Event study of the effect of mugshot bans on the probability that a respondent names crime as the most important problem facing the country. All regressions are weighted by the survey weight. All specifications include state and quarter fixed effects and control for respondents' demographic characteristics and ideology. Panels A and B plot the results for Utah and Louisiana, respectively. The figure shows 95% and 90% confidence intervals based on heteroskedasticity-robust standard errors.

Figure B.21: Event Study. Effect of Mugshot Bans on Crime as the Most Important Problem, Broad Definition

(a) Utah



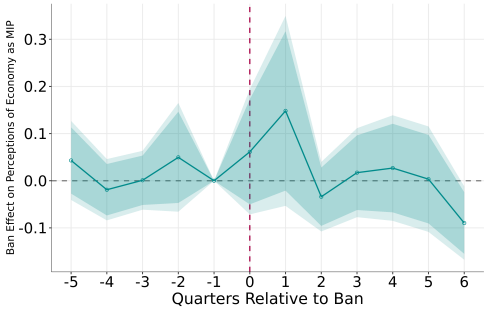
(b) Louisiana



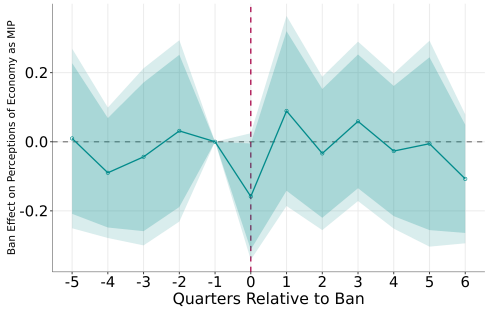
Notes. Event study of the effect of mugshot bans on the probability that a respondent names crime, drugs, the judicial system, police brutality, or school shootings as the most important problem facing the country. All regressions are weighted by the survey weight. All specifications include state and quarter fixed effects and control for respondents' demographic characteristics and ideology. Panels A and B plot the results for Utah and Louisiana, respectively. The figure shows 95% and 90% confidence intervals based on heteroskedasticity-robust standard errors.

Figure B.22: Event Study. Effect of Mugshot Bans on the Economy as the Most Important Problem

(a) Utah



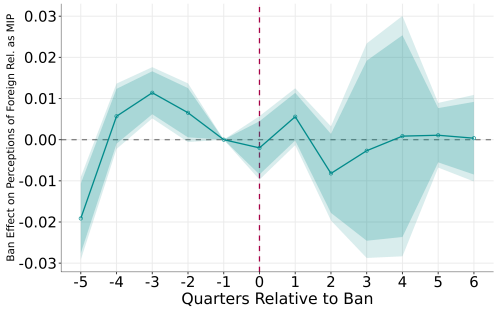
(b) Louisiana



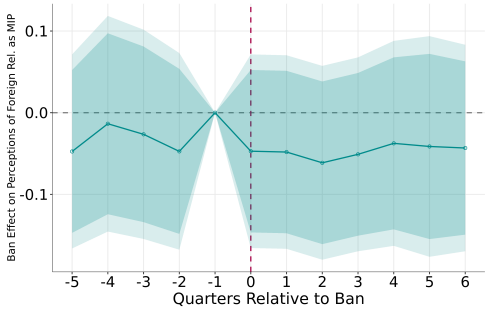
Notes. Event study of the effect of mugshot bans on the probability that a respondent names the economy or unemployment as the most important problem facing the country. All regressions are weighted by the survey weight. All specifications include state and quarter fixed effects and control for respondents' demographic characteristics and ideology. Panels A and B plot the results for Utah and Louisiana, respectively. The figure shows 95% and 90% confidence intervals based on heteroskedasticity-robust standard errors.

Figure B.23: Event Study. Effect of Mugshot Bans on Foreign Affairs as the Most Important Problem

(a) Utah



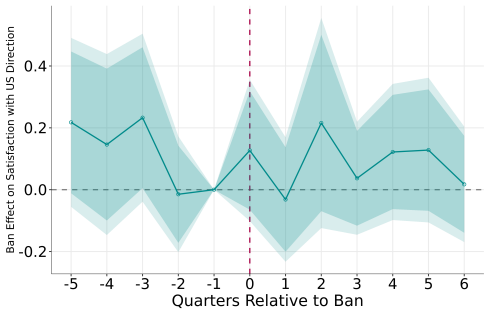
(b) Louisiana



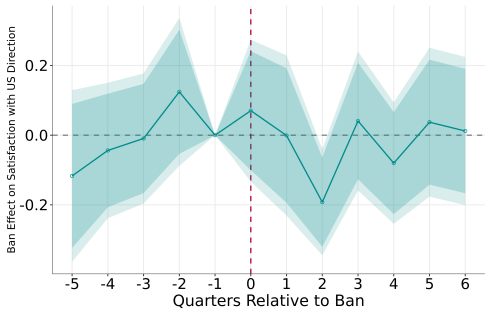
Notes. Event study of the effect of mugshot bans on the probability that a respondent names war, foreign aid, or international affairs as the most important problem facing the country. All regressions are weighted by the survey weight. All specifications include state and quarter fixed effects and control for respondents' demographic characteristics and ideology. Panels A and B plot the results for Utah and Louisiana, respectively. The figure shows 95% and 90% confidence intervals based on heteroskedasticity-robust standard errors.

Figure B.24: Event Study. Effect of Mugshot Bans on Satisfaction with the Direction of the Country

(a) Utah



(b) Louisiana



Notes. Event study of the effect of mugshot bans on the probability that a respondent is satisfied with the way things are going in the United States. All regressions are weighted by the survey weight. All specifications include state and quarter fixed effects and control for respondents' demographic characteristics and ideology. Panels A and B plot the results for Utah and Louisiana, respectively. The figure shows 95% and 90% confidence intervals based on heteroskedasticity-robust standard errors.